

Probabilistic Methods in Stochastic Control

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Preface

This is the reading note of the book *Continuous-time Stochastic Control with Applications in Finance* by Huy  n Pham and the book *Backward Stochastic Differential Equations: From Linear to Fully Nonlinear Theory*. Assuming background knowledge in probability theory and basic stochastic calculus, this note introduces the basic notions of stochastic control and the BSDE approach as the main probabilistic method. To be more specific, this note covers chapters 1, 2, 3, and 6 of the first book and chapters 4, 5, 7, 8 of the second book. In addition, this note is also supplemented by the book *Lectures on BSDEs, Stochastic Control, and Stochastic Differential Games with Financial Applications* by Ren   Carmona (on the section about the stochastic Pontryagin maximum principle). Topics of the note include: preliminaries in stochastic control problems, the classical PDE approach, the BSDE approach, basics of BSDE theory, and two types of special BSDEs: BSDEs with quadratic growth in Z and forward-backward differential equations.

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Chapter 1 Preliminaries

1.1 Continuous-Time Stochastic Processes

We introduce several basic notions for continuous-time stochastic processes.

Definition 1.1.1. A stochastic process is a family

$$X = (X_t)_{t \in \mathbb{T}}$$

of random variables taking values in a measurable space $\mathcal{X}$ and indexed by time t . The time set $\mathbb{T}$ may be discrete or continuous.

Remark. For the purposes of this book, we take $\mathcal{X} = \mathbb{R}^d$, equipped with its Borel σ -field, and consider continuous-time processes. The time interval is either finite, $\mathbb{T} = [0, T]$ with $0 < T < \infty$, or infinite, $\mathbb{T} = [0, \infty)$.

We next introduce filtrations and measurability.

Definition 1.1.2. A filtration on $(\Omega, \mathcal{F}, P)$ is an increasing family

$$\mathbb{F} = (\mathcal{F}_t)_{t \in \mathbb{T}}$$

of sub- σ -fields of $\mathcal{F}$; that is,

$$\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}, \quad 0 \leq s \leq t, \quad s, t \in \mathbb{T}.$$

Definition 1.1.3. Let

$$\mathcal{N}^P := \{N \subseteq A : A \in \mathcal{F}, P(A) = 0\}$$

be the collection of P -negligible sets. A filtration $\mathbb{F}$ satisfies the usual conditions if it is complete,

$$\mathcal{N}^P \subseteq \mathcal{F}_0,$$

and right-continuous. For $t < \sup \mathbb{T}$, right-continuity means

$$\mathcal{F}_{t+} := \bigcap_{\substack{s > t \\ s \in \mathbb{T}}} \mathcal{F}_s = \mathcal{F}_t.$$

On a finite horizon, we use the convention $\mathcal{F}_{T+} = \mathcal{F}_T$.

Remark. Right-continuity means, informally, that observing the filtration immediately after time t reveals no additional information beyond that already available at time t . Completeness means that every subset of a null event is already known at time zero.

Remark. Starting from an arbitrary filtration, one obtains its usual augmentation by adjoining all P -negligible sets and then imposing right-continuity. On an infinite horizon one may write

$$\mathcal{F}_t^{\text{aug}} := \bigcap_{s>t} (\mathcal{F}_s \vee \mathcal{N}^P),$$

and on a finite horizon one uses the analogous definition for $t < T$, together with $\mathcal{F}_T^{\text{aug}} = \mathcal{F}_T \vee \mathcal{N}^P$.

A process can be viewed either pathwise, as a function of time for fixed ω , or as a random variable X_t for fixed t . Considering the joint map $(t, \omega) \mapsto X_t(\omega)$ leads to the following definitions.

Definition 1.1.4 (Progressively measurable, optional, and predictable processes). A process X is *progressively measurable* if, for every $t \in \mathbb{T}$, its restriction to $[0, t] \times \Omega$ is measurable with respect to the product σ -field $\mathcal{B}([0, t]) \otimes \mathcal{F}_t$.

A process is *optional* if $(t, \omega) \mapsto X_t(\omega)$ is measurable with respect to the optional σ -field, namely the σ -field generated by all adapted càdlàg processes.

A process is *predictable* if $(t, \omega) \mapsto X_t(\omega)$ is measurable with respect to the predictable σ -field, namely the σ -field generated by all adapted left-continuous processes.

We shall often study processes through their conditional-expectation behavior.

Definition 1.1.5. An adapted process $X = (X_t)_{t \in \mathbb{T}}$ is a supermartingale if $X_t \in L^1$ for every $t \in \mathbb{T}$ and

$$\mathbb{E}[X_t \mid \mathcal{F}_s] \leq X_s \quad \text{a.s.,} \quad 0 \leq s \leq t, \quad s, t \in \mathbb{T}.$$

It is a submartingale if $-X$ is a supermartingale, and it is a martingale if it is both a supermartingale and a submartingale.

The martingale notion can be localized.

Definition 1.1.6. Let X be an adapted càdlàg process. It is a local martingale if there is an increasing sequence of stopping times $(\tau_n)_{n \geq 1}$ such that $\tau_n \uparrow \infty$ a.s. and each stopped process X^{τ_n} is a martingale. On a finite horizon $[0, T]$, one equivalently requires $\tau_n \uparrow T$ a.s.

Local martingales form the natural setting for stochastic integration. The following criterion is frequently useful for proving that a local martingale is a true martingale.

Proposition 1.1.7. Let $M = (M_t)_{t \in \mathbb{T}}$ be a local martingale. If

$$\mathbb{E} \left[\sup_{0 \leq s \leq t} |M_s| \right] < \infty \quad \text{for every } t \in \mathbb{T},$$

then M is a martingale.

Remark. A càdlàg local martingale is a martingale if and only if it is locally of class (DL): for every deterministic $a < \infty$, the family

$$\{M_\tau : \tau \leq a, \tau \text{ a stopping time}\}$$

is uniformly integrable. On a fixed finite horizon this is the usual class-(D) condition. The preceding proposition is a convenient sufficient criterion.

We now summarize the Snell envelope, which is fundamental in optimal stopping. For $0 \leq t \leq T < \infty$, let $\mathcal{T}_{t,T}$ denote the set of stopping times taking values in $[t, T]$.

Proposition 1.1.8 (Snell envelope). Let $H = (H_t)_{0 \leq t \leq T}$ be a real-valued, adapted, càdlàg process of class (D). Its Snell envelope is

$$V_t := \operatorname{ess\,sup}_{\tau \in \mathcal{T}_{t,T}} \mathbb{E}[H_\tau \mid \mathcal{F}_t], \quad 0 \leq t \leq T.$$

Then V is the smallest càdlàg supermartingale dominating H .

If, in addition, the filtration is continuous and H has no negative jumps,

$$H_t - H_{t-} \geq 0, \quad 0 < t \leq T,$$

then V is continuous and

$$\tau_t^* := \inf\{s \in [t, T] : V_s = H_s\} \wedge T$$

is optimal after time t . In particular,

$$V_t = \mathbb{E}[V_{\tau_t^*} \mid \mathcal{F}_t] = \mathbb{E}[H_{\tau_t^*} \mid \mathcal{F}_t].$$

1.2 Stochastic Calculus

We record several standard tools from stochastic calculus.

Theorem 1.2.1 (Itô's formula). Let X be an n -dimensional Itô process satisfying

$$dX_t = \mu_t dt + \sigma_t dW_t,$$

where $\mu_t \in \mathbb{R}^n$, $\sigma_t \in \mathbb{R}^{n \times m}$, and W is an m -dimensional Brownian motion. If $f \in C^{1,2}([0, T] \times \mathbb{R}^n)$, then

$$\begin{aligned} df(t, X_t) = & \left[\partial_t f(t, X_t) + \nabla_x f(t, X_t)^\top \mu_t + \frac{1}{2} \operatorname{tr}(\sigma_t \sigma_t^\top D_x^2 f(t, X_t)) \right] dt \\ & + \nabla_x f(t, X_t)^\top \sigma_t dW_t. \end{aligned}$$

We next introduce the infinitesimal generator associated with a diffusion.

Definition 1.2.2. Suppose that X solves

$$dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t.$$

For a twice continuously differentiable test function φ , the time-dependent generator is

$$(\mathcal{A}_t \varphi)(x) := \sum_{i=1}^n \mu_i(t, x) \partial_{x_i} \varphi(x) + \frac{1}{2} \sum_{i,j=1}^n (\sigma \sigma^\top)_{ij}(t, x) \partial_{x_i x_j}^2 \varphi(x).$$

For a space-time test function F , the corresponding space-time operator is $\partial_t F + \mathcal{A}_t F$.

The Feynman–Kac formula links diffusion processes with parabolic partial differential equations.

Theorem 1.2.3 (Feynman–Kac formula). Let $X^{t,x}$ be the solution of

$$\begin{aligned} dX_s^{t,x} &= \mu(s, X_s^{t,x}) ds + \sigma(s, X_s^{t,x}) dW_s, \quad t \leq s \leq T, \\ X_t^{t,x} &= x. \end{aligned}$$

Assume that $F \in C^{1,2}([0, T] \times \mathbb{R}^n)$ has sufficient growth control for Itô's formula and solves

$$\begin{aligned} \partial_t F(t, x) + (\mathcal{A}_t F)(t, x) - rF(t, x) &= 0, \quad 0 \leq t < T, \\ F(T, x) &= \Phi(x). \end{aligned}$$

Assume also that

$$\mathbb{E}_{t,x} \left[\int_t^T e^{-2r(s-t)} \left\| \sigma(s, X_s^{t,x})^\top \nabla_x F(s, X_s^{t,x}) \right\|^2 ds \right] < \infty.$$

Then

$$F(t, x) = e^{-r(T-t)} \mathbb{E}_{t,x}[\Phi(X_T^{t,x})],$$

where $\mathbb{E}_{t,x}$ denotes expectation conditional on $X_t = x$.

Transition densities satisfy the Kolmogorov backward and forward equations.

Theorem 1.2.4 (Kolmogorov backward equation). Suppose that the diffusion above has a transition density p defined by

$$P(s, y; t, dx) := P(X_t \in dx \mid X_s = y) = p(s, y; t, x) dx.$$

Under the regularity assumptions needed to differentiate the density,

$$\left(\partial_s + \mathcal{A}_s^{(y)} \right) p(s, y; t, x) = 0, \quad 0 \leq s < t,$$

where $\mathcal{A}_s^{(y)}$ acts on the starting variable y . The terminal condition is

$$p(s, \cdot; t, x) \longrightarrow \delta_x \quad \text{as } s \uparrow t$$

in the sense of distributions in the variable y .

Theorem 1.2.5 (Kolmogorov forward equation). Under the same regularity assumptions, the density also satisfies

$$\partial_t p(s, y; t, x) = \mathcal{A}_t^* p(s, y; t, x), \quad t > s,$$

with initial condition

$$p(s, y; s, \cdot) dx = \delta_y.$$

Writing

$$a(t, x) := \sigma(t, x) \sigma(t, x)^\top,$$

the formal adjoint is

$$(\mathcal{A}_t^* q)(x) = - \sum_{i=1}^n \partial_{x_i} (\mu_i(t, x) q(x)) + \frac{1}{2} \sum_{i,j=1}^n \partial_{x_i x_j}^2 (a_{ij}(t, x) q(x)).$$

This is also called the Fokker–Planck equation.

1.3 Regularity Estimates in Stochastic Analysis

We begin with Doob's maximal inequalities.

Theorem 1.3.1 (Doob's maximal inequalities). Let τ be a bounded stopping time.

(i) If X is a nonnegative càdlàg submartingale, then for every $\lambda > 0$,

$$P\left(\sup_{0 \leq s \leq \tau} X_s \geq \lambda\right) \leq \frac{\mathbb{E}[X_\tau]}{\lambda}.$$

(ii) If X is a càdlàg martingale, $p > 1$, and $X_\tau \in L^p$, then

$$\mathbb{E}\left[\left(\sup_{0 \leq s \leq \tau} |X_s|\right)^p\right] \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}[|X_\tau|^p].$$

Remark. If X is a càdlàg uniformly integrable martingale on an infinite horizon, then the martingale convergence theorem and Doob's inequality imply

$$\sup_{t \geq 0} |X_t| < \infty \quad \text{a.s.}$$

The following inequality controls covariation and is used in stochastic integration.

Proposition 1.3.2 (Kunita–Watanabe inequality). Let M and N be continuous local martingales, and let α and β be predictable processes for which the integrals below are well defined. Then, for every t ,

$$\int_0^t |\alpha_s \beta_s| d|\langle M, N \rangle|_s \leq \left(\int_0^t \alpha_s^2 d\langle M \rangle_s\right)^{1/2} \left(\int_0^t \beta_s^2 d\langle N \rangle_s\right)^{1/2} \quad \text{a.s.}$$

Remark. Set $A := \langle M \rangle + \langle N \rangle$. There are predictable densities a, b, c such that

$$d\langle M \rangle = a dA, \quad d\langle N \rangle = b dA, \quad d\langle M, N \rangle = c dA,$$

and the Kunita–Watanabe inequality yields $|c| \leq \sqrt{ab}$ almost everywhere with respect to dA .

Theorem 1.3.3 (Burkholder–Davis–Gundy inequality). For every $p > 0$, there are constants $c_p, C_p > 0$ such that, for every continuous local martingale M with $M_0 = 0$ and every stopping time τ ,

$$c_p \mathbb{E}[\langle M \rangle_\tau^{p/2}] \leq \mathbb{E} \left[\left(\sup_{0 \leq s \leq \tau} |M_s| \right)^p \right] \leq C_p \mathbb{E}[\langle M \rangle_\tau^{p/2}].$$

The inequalities are understood in the extended sense when one of the expectations is infinite.

Corollary 1.3.4. Let M be a continuous local martingale with $M_0 \in L^1$. If

$$\mathbb{E}[\langle M \rangle_t^{1/2}] < \infty \quad \text{for every finite } t,$$

then M is a martingale.

We next record standard moment and stability estimates for stochastic differential equations.

Theorem 1.3.5 (Existence, uniqueness, and a moment estimate). Consider the $\mathbb{R}^n$ -valued SDE

$$dX_s = b(s, X_s, \omega) ds + \sigma(s, X_s, \omega) dW_s.$$

Assume that, for each x , the coefficient processes are progressively measurable and that there are a deterministic constant $K \geq 0$ and a nonnegative progressively measurable process κ such that

$$\begin{aligned} |b(s, x) - b(s, y)| + \|\sigma(s, x) - \sigma(s, y)\| &\leq K|x - y|, \\ |b(s, x)| + \|\sigma(s, x)\| &\leq \kappa_s + K|x|. \end{aligned}$$

Fix $p \geq 2$ and a finite horizon T , and assume

$$\mathbb{E} \left[\left(\int_0^T \kappa_s^2 ds \right)^{p/2} \right] < \infty.$$

Then, for every $t \leq T$ and every $\mathcal{F}_t$ -measurable $\xi \in L^p(\Omega; \mathbb{R}^n)$, there is a unique strong solution satisfying $X_t = \xi$, and

$$\mathbb{E} \left[\sup_{t \leq s \leq T} |X_s|^p \right] \leq C_{p,T} \left(1 + \mathbb{E}|\xi|^p + \mathbb{E} \left[\left(\int_t^T \kappa_s^2 ds \right)^{p/2} \right] \right).$$

Theorem 1.3.6 (Stability estimates). Under the preceding assumptions, let $X^{t,x}$ denote the solution starting from the deterministic point x at time t . For every finite T , there is a constant C_T

such that, for $t \leq s \leq T$,

$$\mathbb{E} \left[\sup_{t \leq u \leq s} |X_u^{t,x}|^2 \right] \leq C_T \left(|x|^2 + \mathbb{E} \int_t^s \kappa_r^2 \, dr \right),$$

and

$$\mathbb{E} \left[\sup_{t \leq u \leq s} |X_u^{t,x} - x|^2 \right] \leq C_T \mathbb{E} \int_t^s (|x|^2 + \kappa_r^2) \, dr.$$

If, in addition, there is a constant $\beta_0 \in \mathbb{R}$ such that

$$\begin{aligned} & (b(r, x) - b(r, y)) \cdot (x - y) \\ & + \frac{1}{2} \|\sigma(r, x) - \sigma(r, y)\|^2 \leq \beta_0 |x - y|^2 \end{aligned}$$

for all r, x, y , almost surely, then

$$\mathbb{E} |X_s^{t,x} - X_s^{t,y}|^2 \leq e^{2\beta_0(s-t)} |x - y|^2.$$

Moreover, without strengthening the exponential constant to $e^{2\beta_0(s-t)}$, one has the supremum estimate

$$\mathbb{E} \left[\sup_{t \leq u \leq s} |X_u^{t,x} - X_u^{t,y}|^2 \right] \leq C_T |x - y|^2.$$

Proof. Apply Itô's formula to $|X_s|^2$, $|X_s - x|^2$, or $|X_s^{t,x} - X_s^{t,y}|^2$, as appropriate. For estimates involving a supremum, use the Burkholder–Davis–Gundy inequality on the stochastic integral and then Young's inequality to absorb the resulting supremum term. The remaining deterministic integral inequality has the form

$$g(t) \leq h(t) + C \int_0^t g(s) \, ds,$$

so Gronwall's lemma yields the first two estimates and the supremum stability bound. Under the one-sided monotonicity condition, taking expectations before introducing a supremum gives

$$\mathbb{E} |X_s^{t,x} - X_s^{t,y}|^2 \leq |x - y|^2 + 2\beta_0 \int_t^s \mathbb{E} |X_r^{t,x} - X_r^{t,y}|^2 \, dr,$$

and a second application of Gronwall's lemma gives the endpoint estimate. □

Chapter 2 The Classical PDE Approach

2.1 Problem Setup

In this chapter, we use dynamic programming to derive partial differential equations that characterize stochastic control problems.

We first consider a controlled state process in $\mathbb{R}^n$:

$$dX_s = b(s, X_s, \alpha_s) ds + \sigma(s, X_s, \alpha_s) dW_s,$$

where W is a d -dimensional Brownian motion on a filtered probability space

$$(\Omega, \mathcal{F}, \mathbb{F} = (\mathcal{F}_t)_{t \geq 0}, P)$$

satisfying the usual conditions. The control $\alpha = (\alpha_s)$ is progressively measurable and takes values in a measurable set $A \subseteq \mathbb{R}^m$. The coefficient maps

$$b : [0, T] \times \mathbb{R}^n \times A \rightarrow \mathbb{R}^n, \quad \sigma : [0, T] \times \mathbb{R}^n \times A \rightarrow \mathbb{R}^{n \times d}$$

are measurable and uniformly Lipschitz in the state variable: there is a constant $K \geq 0$ such that

$$|b(t, x, a) - b(t, y, a)| + \|\sigma(t, x, a) - \sigma(t, y, a)\| \leq K|x - y|$$

for all $t, x, y \in \mathbb{R}^n$, and $a \in A$. For infinite-horizon problems, we shall normally assume that b and σ are time-homogeneous so that the control problem is stationary.

For $0 \leq t \leq T \leq \infty$, let $\mathcal{T}_{t,T}$ denote the set of stopping times taking values in $[t, T]$. When $t = 0$ and $T = \infty$, we write $\mathcal{T} := \mathcal{T}_{0,\infty}$.

2.1.1 Finite-Horizon Problems

Fix $0 < T < \infty$. Let $\mathcal{A}$ be a class of progressively measurable, A -valued controls such that

$$\mathbb{E} \left[\int_0^T \left(|b(s, 0, \alpha_s)|^2 + \|\sigma(s, 0, \alpha_s)\|^2 \right) ds \right] < \infty.$$

Under the Lipschitz and integrability assumptions, for every $\alpha \in \mathcal{A}$ and every $(t, x) \in [0, T] \times \mathbb{R}^n$, there is a unique strong solution

$$X^{t,x,\alpha} = (X_s^{t,x,\alpha})_{t \leq s \leq T}$$

with $X_t^{t,x,\alpha} = x$. Moreover,

$$\mathbb{E} \left[\sup_{t \leq s \leq T} |X_s^{t,x,\alpha}|^2 \right] < \infty$$

and, whenever $t + h \leq T$,

$$\lim_{h \downarrow 0} \mathbb{E} \left[\sup_{s \in [t, t+h]} |X_s^{t,x,\alpha} - x|^2 \right] = 0.$$

Let

$$f : [0, T] \times \mathbb{R}^n \times A \rightarrow \mathbb{R}, \quad g : \mathbb{R}^n \rightarrow \mathbb{R}$$

be measurable. To keep the value function finite, we assume below that g has at most quadratic growth,

$$|g(x)| \leq C(1 + |x|^2), \quad x \in \mathbb{R}^n.$$

A merely lower-bounded terminal reward can also be allowed if one works with extended-valued expectations, but additional assumptions are then required whenever finiteness is used.

Definition 2.1.1. For $(t, x) \in [0, T] \times \mathbb{R}^n$, let $\mathcal{A}(t, x)$ be the set of controls $\alpha \in \mathcal{A}$ satisfying

$$\mathbb{E} \left[\int_t^T |f(s, X_s^{t,x,\alpha}, \alpha_s)| \, ds \right] < \infty.$$

Assume that $\mathcal{A}(t, x)$ is nonempty. Define

$$J(t, x, \alpha) := \mathbb{E} \left[\int_t^T f(s, X_s^{t,x,\alpha}, \alpha_s) \, ds + g(X_T^{t,x,\alpha}) \right]$$

and

$$v(t, x) := \sup_{\alpha \in \mathcal{A}(t, x)} J(t, x, \alpha).$$

Remark. The measurability of v is not automatic. A rigorous dynamic-programming theorem requires appropriate stability, measurable-selection, and regularity assumptions on the control family and the coefficients.

Definition 2.1.2. A control $\hat{\alpha} \in \mathcal{A}(t, x)$ is optimal at (t, x) if

$$v(t, x) = J(t, x, \hat{\alpha}).$$

A control of the form

$$\alpha_s = a(s, X_s^{t,x,\alpha}),$$

for a measurable map $a : [0, T] \times \mathbb{R}^n \rightarrow A$, is called a Markov control.

2.1.2 Infinite-Horizon Problems

For the stationary infinite-horizon problem, assume that $b = b(x, a)$ and $\sigma = \sigma(x, a)$ are time-homogeneous. Let $\mathcal{A}_0$ be the class of progressively measurable, A -valued controls such that, for

every finite T ,

$$\mathbb{E} \left[\int_0^T \left(|b(0, \alpha_s)|^2 + \|\sigma(0, \alpha_s)\|^2 \right) ds \right] < \infty.$$

For $x \in \mathbb{R}^n$ and $\alpha \in \mathcal{A}_0$, let $X^{x, \alpha}$ be the unique strong solution starting from x at time zero. Standard SDE estimates give, for each finite T ,

$$\mathbb{E} \left[\sup_{0 \leq s \leq T} |X_s^{x, \alpha}|^2 \right] \leq C_T \left(1 + |x|^2 + \mathbb{E} \int_0^T \left(|b(0, \alpha_s)|^2 + \|\sigma(0, \alpha_s)\|^2 \right) ds \right).$$

Fix a discount parameter $\beta > 0$ and define

$$D(t, s) := e^{-\beta(s-t)}, \quad s \geq t.$$

Definition 2.1.3. Let $f : \mathbb{R}^n \times A \rightarrow \mathbb{R}$ be measurable. For $x \in \mathbb{R}^n$, let $\mathcal{A}(x)$ be the set of controls $\alpha \in \mathcal{A}_0$ such that

$$\mathbb{E} \left[\int_0^\infty e^{-\beta s} |f(X_s^{x, \alpha}, \alpha_s)| ds \right] < \infty.$$

Assume that $\mathcal{A}(x)$ is nonempty, and define

$$J(x, \alpha) := \mathbb{E} \left[\int_0^\infty e^{-\beta s} f(X_s^{x, \alpha}, \alpha_s) ds \right], \quad v(x) := \sup_{\alpha \in \mathcal{A}(x)} J(x, \alpha).$$

Definition 2.1.4. A control $\hat{\alpha} \in \mathcal{A}(x)$ is optimal at x if $v(x) = J(x, \hat{\alpha})$. A control of the form $\alpha_s = a(s, X_s^{x, \alpha})$ is Markovian; if a is independent of s , it is called stationary Markovian.

Remark. Time-homogeneous coefficients, a time-independent running reward, and exponential discounting are the structural assumptions that make the infinite-horizon problem stationary and permit a time-independent value function.

2.2 The Dynamic Programming Principle

The dynamic programming principle states that an optimal policy remains optimal after every admissible intermediate time. We state it under the standard hypotheses that the control family is stable under concatenation at stopping times, that the value function is measurable, and that measurable ε -optimal continuation controls can be selected.

Theorem 2.2.1 (Dynamic programming principle). (i) For the finite-horizon problem, let $(t, x) \in [0, T] \times \mathbb{R}^n$ and $\theta \in \mathcal{T}_{t, T}$. Then

$$v(t, x) = \sup_{\alpha \in \mathcal{A}(t, x)} \mathbb{E} \left[\int_t^\theta f(s, X_s^{t, x, \alpha}, \alpha_s) ds + v(\theta, X_\theta^{t, x, \alpha}) \right].$$

(ii) For the infinite-horizon problem, let $x \in \mathbb{R}^n$ and $\theta \in \mathcal{T}$. Then

$$v(x) = \sup_{\alpha \in \mathcal{A}(x)} \mathbb{E} \left[\int_0^\theta e^{-\beta s} f(X_s^{x, \alpha}, \alpha_s) ds + e^{-\beta \theta} v(X_\theta^{x, \alpha}) \right],$$

with the convention $e^{-\beta \theta} = 0$ on $\{\theta = \infty\}$.

Proof. We sketch the finite-horizon case. For a fixed admissible control α , pathwise uniqueness gives the flow property after θ , with the post- θ portion of α shifted in the natural way. Conditioning at θ therefore yields

$$J(t, x, \alpha) = \mathbb{E} \left[\int_t^\theta f(s, X_s^{t,x,\alpha}, \alpha_s) ds + J(\theta, X_\theta^{t,x,\alpha}, \alpha^\theta) \right],$$

where α^θ denotes the continuation control. Since the continuation reward is bounded above by v , taking the supremum over α gives one inequality.

For the reverse inequality, choose a measurable ε -optimal continuation rule for the random state $(\theta, X_\theta^{t,x,\alpha})$, and concatenate it with α at θ . Stability of the admissible class under concatenation makes the pasted control admissible. Its reward is within ε of the right-hand side. Letting $\varepsilon \downarrow 0$ gives the reverse inequality. The infinite-horizon proof is identical after inserting the discount factors. $\square$

2.3 The HJB Equation and Verification Theorems

2.3.1 Derivation of the HJB Equation

We first derive the finite-horizon Hamilton–Jacobi–Bellman equation heuristically. Fix (t, x) , set $\theta = t + h$, and use the constant control $a \in A$ on $[t, t + h]$. The dynamic programming principle gives

$$v(t, x) \geq \mathbb{E} \left[\int_t^{t+h} f(s, X_s^{t,x,a}, a) ds + v(t + h, X_{t+h}^{t,x,a}) \right].$$

For a smooth test function φ , define

$$(\mathcal{L}^a \varphi)(t, x) := b(t, x, a) \cdot D_x \varphi(t, x) + \frac{1}{2} \text{tr} \left(\sigma(t, x, a) \sigma(t, x, a)^\top D_x^2 \varphi(t, x) \right).$$

Assuming $v \in C^{1,2}$, Itô's formula gives

$$\begin{aligned} v(t + h, X_{t+h}^{t,x,a}) &= v(t, x) + \int_t^{t+h} (\partial_t v + \mathcal{L}^a v)(s, X_s^{t,x,a}) ds \\ &\quad + \int_t^{t+h} D_x v(s, X_s^{t,x,a})^\top \sigma(s, X_s^{t,x,a}, a) dW_s. \end{aligned}$$

After localization, or under a suitable growth condition, the stochastic integral has zero expectation. Hence

$$0 \geq \mathbb{E} \left[\int_t^{t+h} \{ (\partial_t v + \mathcal{L}^a v)(s, X_s^{t,x,a}) + f(s, X_s^{t,x,a}, a) \} ds \right].$$

Dividing by h , then using continuity and dominated convergence as $h \downarrow 0$, gives

$$0 \geq \partial_t v(t, x) + \mathcal{L}^a v(t, x) + f(t, x, a).$$

Since this holds for every $a \in A$,

$$-\partial_t v(t, x) - \sup_{a \in A} \{ \mathcal{L}^a v(t, x) + f(t, x, a) \} \geq 0.$$

If an optimal feedback selector exists and attains the supremum, equality holds. Thus the formal HJB equation is

$$-\partial_t v(t, x) - \sup_{a \in A} \{ \mathcal{L}^a v(t, x) + f(t, x, a) \} = 0, \quad (t, x) \in [0, T) \times \mathbb{R}^n,$$

with terminal condition

$$v(T, x) = g(x).$$

Let

$$\mathcal{S}_n := \{ M \in \mathbb{R}^{n \times n} : M^\top = M \}.$$

For $(t, x, p, M) \in [0, T] \times \mathbb{R}^n \times \mathbb{R}^n \times \mathcal{S}_n$, define the Hamiltonian

$$H(t, x, p, M) := \sup_{a \in A} \left\{ b(t, x, a) \cdot p + \frac{1}{2} \text{tr} \left(\sigma(t, x, a) \sigma(t, x, a)^\top M \right) + f(t, x, a) \right\}.$$

Then the HJB equation becomes

$$-\partial_t v(t, x) - H(t, x, D_x v(t, x), D_x^2 v(t, x)) = 0.$$

Remark. Suppose that the supremum is attained and that there is a measurable selector

$$a^*(t, x) \in \arg \max_{a \in A} \{ \mathcal{L}^a v(t, x) + f(t, x, a) \}.$$

If the feedback SDE

$$\begin{aligned} dX_s^* &= b(s, X_s^*, a^*(s, X_s^*)) ds + \sigma(s, X_s^*, a^*(s, X_s^*)) dW_s, \\ X_t^* &= x, \end{aligned}$$

has a unique admissible solution, then the linear PDE along the selected feedback and the Feynman–Kac formula give

$$v(t, x) = \mathbb{E} \left[\int_t^T f(s, X_s^*, a^*(s, X_s^*)) ds + g(X_T^*) \right].$$

The feedback process $a^*(s, X_s^*)$ is therefore an optimal Markov control.

Remark. For the stationary infinite-horizon problem, the formal HJB equation is

$$\beta v(x) - \sup_{a \in A} \{ \mathcal{L}^a v(x) + f(x, a) \} = 0, \quad x \in \mathbb{R}^n,$$

where

$$(\mathcal{L}^a \varphi)(x) = b(x, a) \cdot D\varphi(x) + \frac{1}{2} \text{tr} \left(\sigma(x, a) \sigma(x, a)^\top D^2 \varphi(x) \right).$$

Remark. A finite-horizon problem may also contain a state- and control-dependent discount factor

$$\Gamma(t, s) := \exp \left(- \int_t^s \beta(u, X_u^{t, x, \alpha}, \alpha_u) du \right).$$

Then

$$J(t, x, \alpha) = \mathbb{E} \left[\int_t^T \Gamma(t, s) f(s, X_s^{t, x, \alpha}, \alpha_s) ds + \Gamma(t, T) g(X_T^{t, x, \alpha}) \right],$$

and the Hamiltonian becomes

$$H(t, x, r, p, M) := \sup_{a \in A} \left\{ -\beta(t, x, a)r + b(t, x, a) \cdot p + \frac{1}{2} \text{tr} \left(\sigma(t, x, a) \sigma(t, x, a)^\top M \right) + f(t, x, a) \right\}.$$

2.3.2 Verification Theorems

The classical verification method shows that a sufficiently regular supersolution bounds the value function and that a smooth solution, together with an admissible maximizing feedback, is the value function.

Theorem 2.3.1 (Finite-horizon supersolution verification bound). Let

$$w \in C^{1,2}([0, T] \times \mathbb{R}^n) \cap C([0, T] \times \mathbb{R}^n).$$

Assume that there are constants $C > 0$ and $p \geq 1$ such that

$$|w(t, x)| \leq C(1 + |x|^p),$$

and assume that all admissible state processes have the corresponding p -th moments needed below. Suppose

$$\begin{aligned} -\partial_t w(t, x) - \sup_{a \in A} \{ \mathcal{L}^a w(t, x) + f(t, x, a) \} &\geq 0, \\ w(T, x) &\geq g(x). \end{aligned}$$

Then

$$w(t, x) \geq v(t, x) \quad \text{on } [0, T] \times \mathbb{R}^n.$$

Proof. Fix (t, x) and $\alpha \in \mathcal{A}(t, x)$, and abbreviate $X = X^{t, x, \alpha}$. Let

$$\tau_n := \inf \left\{ s \geq t : \int_t^s \left\| D_x w(u, X_u)^\top \sigma(u, X_u, \alpha_u) \right\|^2 du \geq n \right\}.$$

For $s < T$, Itô's formula gives

$$\begin{aligned} w(s \wedge \tau_n, X_{s \wedge \tau_n}) &= w(t, x) + \int_t^{s \wedge \tau_n} (\partial_t w + \mathcal{L}^{\alpha_u} w)(u, X_u) du \\ &\quad + \int_t^{s \wedge \tau_n} D_x w(u, X_u)^\top \sigma(u, X_u, \alpha_u) dW_u. \end{aligned}$$

The stopped stochastic integral is a martingale. Since the HJB supersolution inequality implies

$$(\partial_t w + \mathcal{L}^{\alpha_u} w)(u, X_u) + f(u, X_u, \alpha_u) \leq 0,$$

taking expectations yields

$$\mathbb{E}[w(s \wedge \tau_n, X_{s \wedge \tau_n})] \leq w(t, x) - \mathbb{E} \left[\int_t^{s \wedge \tau_n} f(u, X_u, \alpha_u) du \right].$$

The polynomial growth and moment assumptions make the stopped terminal terms uniformly integrable, while the definition of $\mathcal{A}(t, x)$ controls the running reward. Hence one may first let $n \rightarrow \infty$, then $s \uparrow T$, to obtain

$$\mathbb{E}[g(X_T)] \leq w(t, x) - \mathbb{E} \left[\int_t^T f(u, X_u, \alpha_u) du \right].$$

Thus $w(t, x) \geq J(t, x, \alpha)$. Taking the supremum over α proves the claim. $\square$

Theorem 2.3.2 (Finite-horizon verification theorem). In addition to the assumptions of the preceding theorem, suppose that

$$w(T, \cdot) = g$$

and that there is a measurable selector $\hat{a} : [0, T) \times \mathbb{R}^n \rightarrow A$ satisfying

$$\begin{aligned} 0 &= -\partial_t w(t, x) - \sup_{a \in A} \{ \mathcal{L}^a w(t, x) + f(t, x, a) \} \\ &= -\partial_t w(t, x) - \mathcal{L}^{\hat{a}(t, x)} w(t, x) - f(t, x, \hat{a}(t, x)). \end{aligned}$$

Assume that the feedback SDE

$$d\hat{X}_s = b(s, \hat{X}_s, \hat{a}(s, \hat{X}_s)) ds + \sigma(s, \hat{X}_s, \hat{a}(s, \hat{X}_s)) dW_s, \quad \hat{X}_t = x,$$

has a unique solution and that $(\hat{a}(s, \hat{X}_s))_{t \leq s \leq T}$ is admissible. Then

$$w = v,$$

and the feedback control $\hat{a}(s, \hat{X}_s)$ is optimal.

Proof. The preceding theorem gives $w \geq v$. Applying the same localized Itô argument to $w(s, \hat{X}_s)$, now using equality in the HJB equation, gives

$$w(t, x) = \mathbb{E} \left[\int_t^T f(u, \hat{X}_u, \hat{a}(u, \hat{X}_u)) du + g(\hat{X}_T) \right].$$

The right-hand side is bounded above by $v(t, x)$, so $w = v$ and the feedback is optimal. $\square$

Infinite-horizon verification requires a transversality condition.

Theorem 2.3.3 (Infinite-horizon supersolution verification bound). Let $w \in C^2(\mathbb{R}^n)$ have polynomial growth and suppose

$$\beta w(x) - \sup_{a \in A} \{ \mathcal{L}^a w(x) + f(x, a) \} \geq 0.$$

Assume that, for every x and every $\alpha \in \mathcal{A}(x)$,

$$\limsup_{T \rightarrow \infty} e^{-\beta T} \mathbb{E}[w(X_T^{x, \alpha})] \geq 0,$$

and that the family of discounted terminal terms is integrable enough for the localization limits below. Then $w \geq v$ on $\mathbb{R}^n$.

Proof. Fix $\alpha \in \mathcal{A}(x)$. Applying Itô's formula to $e^{-\beta t} w(X_t^{x,\alpha})$, localizing the stochastic integral, and then removing the localization gives

$$w(x) \geq \mathbb{E} \left[\int_0^T e^{-\beta u} f(X_u^{x,\alpha}, \alpha_u) \, du \right] + e^{-\beta T} \mathbb{E}[w(X_T^{x,\alpha})].$$

The discounted running reward converges in L^1 as $T \rightarrow \infty$. Taking the limsup and using the transversality condition therefore yields

$$w(x) \geq J(x, \alpha).$$

Taking the supremum over α proves $w \geq v$. □

Remark. The transversality condition does not assert that the discounted terminal term converges to zero. It ensures only that its asymptotic contribution cannot lower the supersolution bound.

Theorem 2.3.4 (Infinite-horizon verification theorem). Assume the hypotheses of the preceding theorem. Suppose there is a measurable stationary selector $\hat{a} : \mathbb{R}^n \rightarrow A$ such that

$$\begin{aligned} 0 &= \beta w(x) - \sup_{a \in A} \{ \mathcal{L}^a w(x) + f(x, a) \} \\ &= \beta w(x) - \mathcal{L}^{\hat{a}(x)} w(x) - f(x, \hat{a}(x)). \end{aligned}$$

Assume that the feedback SDE has a unique solution $\hat{X}^x$, that $\hat{a}(\hat{X}_s^x)$ is admissible, and that

$$\liminf_{T \rightarrow \infty} e^{-\beta T} \mathbb{E}[w(\hat{X}_T^x)] \leq 0.$$

Then

$$w(x) = v(x) = J(x, \hat{a}),$$

and $\hat{a}$ is an optimal stationary Markov control.

Proof. Equality in the HJB equation gives, for every finite T ,

$$w(x) = \mathbb{E} \left[\int_0^T e^{-\beta u} f(\hat{X}_u^x, \hat{a}(\hat{X}_u^x)) \, du \right] + e^{-\beta T} \mathbb{E}[w(\hat{X}_T^x)].$$

Taking the liminf and using the transversality condition gives $w(x) \leq J(x, \hat{a})$. The supersolution theorem gives $w \geq v \geq J(x, \hat{a})$, so equality holds throughout. □

2.4 Applications

2.4.1 Portfolio Optimization with Quadratic Transaction Costs

Consider an investor whose expected excess-return signal follows an Ornstein–Uhlenbeck process

$$d\mu_s = -\ell\mu_s ds + \eta dW_s.$$

Let q_s denote the risky position and let the trading rate u_s be the control:

$$dq_s = u_s ds.$$

For current state $(\mu_t, q_t) = (m, q)$, define

$$V(m, q) := \sup_u \mathbb{E}_t \left[\int_t^\infty e^{-\beta(s-t)} \left(\mu_s q_s - \frac{\gamma\sigma^2}{2} q_s^2 - \frac{\lambda}{2} u_s^2 \right) ds \right].$$

The stationary HJB equation is

$$0 = -\beta V + \sup_{u \in \mathbb{R}} \left\{ mq - \frac{\gamma\sigma^2}{2} q^2 - \frac{\lambda}{2} u^2 + \mathcal{L}^\mu V + V_q u \right\},$$

where

$$\mathcal{L}^\mu V(m, q) = -\ell m V_m(m, q) + \frac{\eta^2}{2} V_{mm}(m, q).$$

Pointwise optimization gives

$$u^*(m, q) = \frac{V_q(m, q)}{\lambda}$$

and therefore

$$0 = -\beta V + mq - \frac{\gamma\sigma^2}{2} q^2 - \ell m V_m + \frac{\eta^2}{2} V_{mm} + \frac{(V_q)^2}{2\lambda}.$$

Because the running reward is quadratic in (m, q, u) and the state dynamics are linear, it is natural to seek a quadratic value function of the form

$$V(m, q) = \frac{A}{2} m^2 + Bmq - \frac{C}{2} q^2 + D.$$

Substitution produces four algebraic equations for A, B, C, D .

2.4.2 A Singular Stochastic Control Problem

Consider

$$dX_s = \alpha_s dW_s, \quad X_t = x,$$

where α is progressively measurable and satisfies

$$\mathbb{E} \int_t^T |\alpha_s|^2 ds < \infty.$$

Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be measurable, have at most linear growth, and possess a finite concave envelope g^{conc} . Define

$$v(t, x) := \sup_{\alpha} \mathbb{E}[g(X_T^{t, x, \alpha})].$$

The dynamic programming principle gives, for a constant control a and a bounded stopping time $\theta \in \mathcal{T}_{t, T}$,

$$v(t, x) \geq \mathbb{E}[v(\theta, X_{\theta}^{t, x, a})].$$

If $v \in C^{1,2}$, applying Itô's formula up to

$$\theta = (t + h) \wedge \inf\{s \geq t : |X_s^{t, x, a} - x| \geq 1\}$$

gives

$$0 \geq \mathbb{E} \left[\frac{1}{h} \int_t^{\theta} \left(\partial_t v + \frac{1}{2} a^2 \partial_{xx} v \right) (s, X_s^{t, x, a}) ds \right].$$

Letting $h \downarrow 0$,

$$0 \geq \partial_t v(t, x) + \frac{1}{2} a^2 \partial_{xx} v(t, x) \quad \text{for every } a \in \mathbb{R}.$$

Consequently, any classical solution must satisfy $\partial_{xx} v \leq 0$, so $v(t, \cdot)$ is concave for $t < T$. Since the zero control is admissible, $v \geq g$, and hence any such concave value function must satisfy

$$v(t, x) \geq g^{\text{conc}}(x).$$

Conversely, Jensen's inequality and the martingale property of $X^{t, x, \alpha}$ give

$$\begin{aligned} v(t, x) &\leq \sup_{\alpha} \mathbb{E}[g^{\text{conc}}(X_T^{t, x, \alpha})] \\ &\leq \sup_{\alpha} g^{\text{conc}}(\mathbb{E}[X_T^{t, x, \alpha}]) = g^{\text{conc}}(x). \end{aligned}$$

Thus the formal conclusion is

$$v(t, x) = g^{\text{conc}}(x), \quad t < T.$$

This identity is rigorous under standard martingale-representation assumptions or in the viscosity-solution formulation.

For example, take

$$g(x) = -|x|.$$

Then g is concave but not C^2 , and Jensen's inequality together with the zero control gives

$$v(t, x) = -|x|.$$

Hence the value function is not a classical $C^{1,2}$ solution.

Terminal discontinuity can also occur. If

$$g_0(x) := -|x| - \mathbf{1}_{\{0\}}(x),$$

then $g_0^{\text{conc}}(x) = -|x|$. For $t < T$, the value equals $-|x|$, while $v(T, 0) = g_0(0) = -1$; therefore v is discontinuous at the terminal point $(T, 0)$.

The Hamiltonian for this problem is

$$H(M) = \sup_{a \in \mathbb{R}} \frac{1}{2} a^2 M.$$

It is finite exactly when $M \leq 0$, in which case $H(M) = 0$. The corresponding HJB equation is therefore represented by the variational inequality

$$\min\{-\partial_t v, -\partial_{xx} v\} = 0,$$

which is naturally interpreted in the viscosity sense when the concave envelope is nonsmooth.

Chapter 3 The BSDE Approach

3.1 The Stochastic Pontryagin Maximum Principle

Consider the controlled state equation

$$dX_t = b(t, X_t, \alpha_t) dt + \sigma(t, X_t, \alpha_t) dW_t, \quad X_0 = x,$$

where $X_t \in \mathbb{R}^n$, $\alpha_t \in A \subseteq \mathbb{R}^k$, and W is a d -dimensional Brownian motion. The objective is to maximize

$$J(\alpha) := \mathbb{E} \left[\int_0^T f(t, X_t, \alpha_t) dt + g(X_T) \right].$$

Assume that, for almost every (t, ω) , the functions b , σ , and f are continuously differentiable in (x, a) , with bounded Lipschitz derivatives, and that g is continuously differentiable with bounded Lipschitz derivative. If the coefficients are random fields, assume in addition the progressive measurability needed for the state equation and the stochastic integrals. Let A be convex, and assume that the admissible-control set $\mathbb{A}$ is stable under convex combinations.

3.1.1 First-Order Necessary Conditions

At a constrained maximizer, a feasible one-sided directional derivative is nonpositive; it need not vanish. We first compute the variation of the state process.

Fix $\alpha \in \mathbb{A}$, let $X = X^\alpha$, choose another control $\alpha' \in \mathbb{A}$, and set

$$\beta := \alpha' - \alpha.$$

For $0 \leq \varepsilon \leq 1$, define

$$\alpha^\varepsilon := (1 - \varepsilon)\alpha + \varepsilon\alpha' = \alpha + \varepsilon\beta,$$

and let $X^\varepsilon = X^{\alpha^\varepsilon}$. We assume the fourth-moment integrability needed below, for example

$$\mathbb{E} \left[\left(\int_0^T |\beta_t|^2 dt \right)^2 \right] < \infty.$$

Define the first-variation process V by the linear SDE

$$\begin{aligned} dV_t &= [D_x b(t, X_t, \alpha_t) V_t + D_a b(t, X_t, \alpha_t) \beta_t] dt \\ &\quad + [D_x \sigma(t, X_t, \alpha_t) V_t + D_a \sigma(t, X_t, \alpha_t) \beta_t] dW_t, \quad V_0 = 0. \end{aligned}$$

The boundedness of the coefficient derivatives and the moment assumption on β imply the corresponding moment estimates for V .

Lemma 3.1.1 (First variation of the state). Under the preceding assumptions,

$$\lim_{\varepsilon \downarrow 0} \mathbb{E} \left[\sup_{0 \leq t \leq T} \left| \frac{X_t^\varepsilon - X_t}{\varepsilon} - V_t \right|^2 \right] = 0.$$

Proof. Set

$$R_t^\varepsilon := \frac{X_t^\varepsilon - X_t}{\varepsilon} - V_t.$$

By the fundamental theorem of calculus in the variables (x, a) , the drift of R^ε can be written as

$$B_t^\varepsilon R_t^\varepsilon + r_t^{b,\varepsilon},$$

and its diffusion coefficient as

$$\Sigma_t^\varepsilon R_t^\varepsilon + r_t^{\sigma,\varepsilon},$$

where B^ε and Σ^ε are uniformly bounded. The Lipschitz continuity of the first derivatives, together with uniform fourth-moment estimates for X^ε , V , and β , gives

$$\mathbb{E} \int_0^T \left(|r_t^{b,\varepsilon}|^2 + \|r_t^{\sigma,\varepsilon}\|^2 \right) dt \longrightarrow 0.$$

The Burkholder–Davis–Gundy and Young inequalities therefore yield, for $0 \leq s \leq T$,

$$\mathbb{E} \left[\sup_{0 \leq u \leq s} |R_u^\varepsilon|^2 \right] \leq C \int_0^s \mathbb{E} \left[\sup_{0 \leq u \leq r} |R_u^\varepsilon|^2 \right] dr + \delta_\varepsilon,$$

where $\delta_\varepsilon \rightarrow 0$. Gronwall's lemma completes the proof. $\square$

We can now differentiate the objective in feasible directions.

Proposition 3.1.2 (Directional derivative of the objective). For every feasible direction β , the one-sided directional derivative exists and is given by

$$\begin{aligned} DJ(\alpha; \beta) &:= \lim_{\varepsilon \downarrow 0} \frac{J(\alpha + \varepsilon \beta) - J(\alpha)}{\varepsilon} \\ &= \mathbb{E} \left[\int_0^T (D_x f(t, X_t, \alpha_t) \cdot V_t + D_a f(t, X_t, \alpha_t) \cdot \beta_t) dt + Dg(X_T) \cdot V_T \right]. \end{aligned}$$

If the admissible domain is open in a linear space and the same formula holds in both directions, this is the Gâteaux derivative.

Proof. Write

$$X_t^\varepsilon = X_t + \varepsilon(V_t + R_t^\varepsilon).$$

Apply the fundamental theorem of calculus to $f(t, X_t^\varepsilon, \alpha_t^\varepsilon) - f(t, X_t, \alpha_t)$ and to $g(X_T^\varepsilon) - g(X_T)$.

The first-variation lemma, the boundedness of the derivatives, and the dominated convergence

theorem give the stated limit. In the terminal term the perturbation is $V_T + R_T^\varepsilon$, not a process evaluated at an unrelated time. $\square$

3.1.2 The Adjoint Equation and Adjoint Processes

Definition 3.1.3. For

$$(t, x, a, y, z) \in [0, T] \times \mathbb{R}^n \times A \times \mathbb{R}^n \times \mathbb{R}^{n \times d},$$

define the Hamiltonian

$$H(t, x, a, y, z) := b(t, x, a)^\top y + \text{tr}(\sigma(t, x, a)^\top z) + f(t, x, a).$$

Given an admissible control α and its state process X , the associated adjoint processes are a solution (Y, Z) of

$$\begin{aligned} dY_t &= -D_x H(t, X_t, \alpha_t, Y_t, Z_t) dt + Z_t dW_t, \\ Y_T &= Dg(X_T). \end{aligned}$$

Lemma 3.1.4 (Duality relation). Let V be the first-variation process in the direction β . Then

$$\begin{aligned} \mathbb{E}[Y_T^\top V_T] &= \mathbb{E} \int_0^T \left[Y_t^\top D_a b(t, X_t, \alpha_t) \beta_t - D_x f(t, X_t, \alpha_t) \cdot V_t \right. \\ &\quad \left. + \text{tr}(Z_t^\top D_a \sigma(t, X_t, \alpha_t) [\beta_t]) \right] dt. \end{aligned}$$

Proof. Apply Itô's product formula to $Y_t^\top V_t$. The terms containing $D_x b$ and $D_x \sigma$ cancel against the corresponding terms in $D_x H$. After localization, the stochastic integrals have zero expectation, which gives the identity. $\square$

Corollary 3.1.5. The directional derivative can be written solely in terms of the Hamiltonian:

$$DJ(\alpha; \beta) = \mathbb{E} \int_0^T D_a H(t, X_t, \alpha_t, Y_t, Z_t) \cdot \beta_t dt.$$

The sign is positive because the objective is being maximized and the Hamiltonian includes the running reward with a positive sign.

Theorem 3.1.6 (First-order maximum principle). Let $\hat{\alpha} \in \mathcal{A}$ be optimal, with associated state $\hat{X}$ and adjoint processes $(\hat{Y}, \hat{Z})$. Assume that the admissible set permits the usual spike variations. Then, for almost every t , almost surely,

$$D_a H(t, \hat{X}_t, \hat{\alpha}_t, \hat{Y}_t, \hat{Z}_t) \cdot (a - \hat{\alpha}_t) \leq 0, \quad a \in A.$$

Proof. For every admissible α' , optimality and convexity give

$$DJ(\hat{\alpha}; \alpha' - \hat{\alpha}) \leq 0.$$

Using the Hamiltonian representation of the directional derivative and testing with spike varia-

tions on short time intervals gives the pointwise inequality. $\square$

Remark. If $a \mapsto H(t, \hat{X}_t, a, \hat{Y}_t, \hat{Z}_t)$ is concave, the first-order condition is equivalent to the global maximum condition

$$H(t, \hat{X}_t, \hat{a}_t, \hat{Y}_t, \hat{Z}_t) \geq H(t, \hat{X}_t, a, \hat{Y}_t, \hat{Z}_t), \quad a \in A.$$

Without concavity, the first-order inequality need not characterize a global maximizer.

3.1.3 Sufficient Conditions for Optimality

Theorem 3.1.7 (Sufficient stochastic maximum principle). Let $\hat{a} \in \mathbb{A}$, with associated state $\hat{X}$ and adjoint processes $(\hat{Y}, \hat{Z})$. Assume that

- (i) g is concave;
- (ii) for almost every t , almost surely, the map

$$(x, a) \mapsto H(t, x, a, \hat{Y}_t, \hat{Z}_t)$$

is jointly concave;

- (iii) for almost every t , almost surely,

$$H(t, \hat{X}_t, \hat{a}_t, \hat{Y}_t, \hat{Z}_t) = \sup_{a \in A} H(t, \hat{X}_t, a, \hat{Y}_t, \hat{Z}_t).$$

Then $\hat{a}$ is optimal:

$$J(\hat{a}) = \sup_{\alpha \in \mathbb{A}} J(\alpha).$$

Proof. Fix $\alpha \in \mathbb{A}$, and let X be its state process. Concavity of g gives

$$\mathbb{E}[g(\hat{X}_T) - g(X_T)] \geq \mathbb{E}[\hat{Y}_T^\top (\hat{X}_T - X_T)].$$

Applying Itô's product formula to $\hat{Y}_t^\top (\hat{X}_t - X_t)$, taking expectations, and using the adjoint equation yields

$$\begin{aligned} J(\hat{a}) - J(\alpha) &\geq \mathbb{E} \int_0^T \left[H(t, \hat{X}_t, \hat{a}_t, \hat{Y}_t, \hat{Z}_t) - H(t, X_t, \alpha_t, \hat{Y}_t, \hat{Z}_t) \right. \\ &\quad \left. - D_x H(t, \hat{X}_t, \hat{a}_t, \hat{Y}_t, \hat{Z}_t) \cdot (\hat{X}_t - X_t) \right] dt. \end{aligned}$$

Joint concavity and the maximum condition imply that the integrand is nonnegative. Hence

$$J(\hat{a}) \geq J(\alpha). \quad \square$$

3.1.4 Connection with Dynamic Programming

The HJB and maximum-principle approaches are linked when the value function is smooth. Define

$$\mathcal{G}(t, x, a, p, M) := b(t, x, a) \cdot p + \frac{1}{2} \text{tr} \left(\sigma(t, x, a) \sigma(t, x, a)^\top M \right) + f(t, x, a).$$

The HJB equation is

$$-\partial_t v(t, x) - \sup_{a \in A} \mathcal{G}(t, x, a, D_x v(t, x), D_x^2 v(t, x)) = 0, \quad v(T, \cdot) = g.$$

Theorem 3.1.8 (Adjoint processes from the value function). Assume that

$$v \in C^{1,3}([0, T] \times \mathbb{R}^n) \cap C([0, T] \times \mathbb{R}^n)$$

solves the HJB equation. Suppose there is a measurable feedback selector $\hat{a}(t, x)$ attaining the supremum, with enough regularity for the envelope differentiation below, and let

$$\hat{a}_t := \hat{a}(t, \hat{X}_t)$$

be an optimal admissible feedback control. Then

$$\hat{Y}_t = D_x v(t, \hat{X}_t), \quad \hat{Z}_t = D_x^2 v(t, \hat{X}_t) \sigma(t, \hat{X}_t, \hat{a}_t)$$

solve the adjoint BSDE.

Proof. Along the optimal feedback, the HJB equation reads

$$\partial_t v(t, x) + \mathcal{G}(t, x, \hat{a}(t, x), D_x v(t, x), D_x^2 v(t, x)) = 0.$$

Differentiating with respect to x_i and using the envelope condition gives, along $x = \hat{X}_t$,

$$\begin{aligned} 0 = & \partial_{tx_i}^2 v + \sum_{j=1}^n b_j \partial_{x_j x_i}^2 v + \frac{1}{2} \sum_{j,k=1}^n (\sigma \sigma^\top)_{jk} \partial_{x_j x_k x_i}^3 v \\ & + \partial_{x_i} H(t, \hat{X}_t, \hat{a}_t, D_x v, D_x^2 v \sigma), \end{aligned}$$

where all coefficients and derivatives of v on the right-hand side are evaluated at $(t, \hat{X}_t)$. This componentwise formula is the precise meaning of the contraction between $\sigma \sigma^\top$ and the third derivative of v .

Applying Itô's formula componentwise to $D_x v(t, \hat{X}_t)$ now gives

$$d\hat{Y}_t = -D_x H(t, \hat{X}_t, \hat{a}_t, \hat{Y}_t, \hat{Z}_t) dt + \hat{Z}_t dW_t.$$

Finally, $v(T, \cdot) = g$ implies $\hat{Y}_T = Dg(\hat{X}_T)$. □

3.2 Applications

3.2.1 Mean–Variance Portfolio Optimization

Consider a Black–Scholes market with a risk-free asset

$$dS_t^0 = rS_t^0 dt$$

and a risky asset

$$dS_t = S_t(b dt + \sigma dW_t),$$

where $b > r$ and $\sigma > 0$. At time t , an investor places the amount α_t in the risky asset. The wealth process satisfies

$$\begin{aligned} dX_t &= \alpha_t \frac{dS_t}{S_t} + (X_t - \alpha_t) \frac{dS_t^0}{S_t^0} \\ &= [rX_t + \alpha_t(b - r)] dt + \sigma \alpha_t dW_t, \quad X_0 = x. \end{aligned}$$

Let $\mathcal{A}$ be the class of progressively measurable controls satisfying

$$\mathbb{E} \int_0^T |\alpha_t|^2 dt < \infty.$$

For a target mean m , the mean–variance problem is

$$V(m) := \inf_{\alpha \in \mathcal{A}} \{ \text{Var}(X_T) : \mathbb{E}[X_T] = m \}.$$

The associated auxiliary problem is

$$\tilde{V}(\lambda) := \inf_{\alpha \in \mathcal{A}} \mathbb{E}[(X_T - \lambda)^2], \quad \lambda \in \mathbb{R}.$$

This is a minimization problem, so the maximum-principle sign convention is replaced by the corresponding minimum condition. Its Hamiltonian is

$$H(x, a, y, z) = [rx + a(b - r)]y + \sigma az,$$

and the adjoint equation is

$$-dY_t = rY_t dt - Z_t dW_t, \quad Y_T = 2(X_T - \lambda).$$

Let $\hat{\alpha}$ be a candidate optimum. Since the Hamiltonian is linear in a , the first-order condition is

$$(b - r)\hat{Y}_t + \sigma \hat{Z}_t = 0.$$

Seek the adjoint process in the form

$$\hat{Y}_t = \varphi(t)\hat{X}_t + \psi(t).$$

Then

$$\hat{Z}_t = \varphi(t)\sigma\hat{\alpha}_t$$

and the terminal conditions are

$$\varphi(T) = 2, \quad \psi(T) = -2\lambda.$$

The first-order condition gives

$$\hat{\alpha}_t = \frac{(r-b)(\varphi(t)\hat{X}_t + \psi(t))}{\sigma^2\varphi(t)}.$$

Matching the drift terms yields

$$\begin{aligned} \varphi'(t) + \left(2r - \frac{(b-r)^2}{\sigma^2}\right) \varphi(t) &= 0, \quad \varphi(T) = 2, \\ \psi'(t) + \left(r - \frac{(b-r)^2}{\sigma^2}\right) \psi(t) &= 0, \quad \psi(T) = -2\lambda. \end{aligned}$$

Hence

$$\begin{aligned} \varphi(t) &= 2 \exp\left[\left(2r - \frac{(b-r)^2}{\sigma^2}\right)(T-t)\right], \\ \psi_\lambda(t) &= -2\lambda \exp\left[\left(r - \frac{(b-r)^2}{\sigma^2}\right)(T-t)\right]. \end{aligned}$$

The optimal feedback for the auxiliary problem is therefore

$$\hat{\alpha}_\lambda(t, x) = \frac{(r-b)(\varphi(t)x + \psi_\lambda(t))}{\sigma^2\varphi(t)}.$$

Completing the square after applying Itô's formula to $\frac{1}{2}\varphi(t)X_t^2 + \psi_\lambda(t)X_t$ gives

$$\begin{aligned} \mathbb{E}[(X_T - \lambda)^2] &= \frac{1}{2}\varphi(0)x^2 + \psi_\lambda(0)x + \lambda^2 \\ &\quad + \mathbb{E} \int_0^T \frac{\varphi(t)\sigma^2}{2} \left(\alpha_t - \frac{(r-b)(\varphi(t)X_t + \psi_\lambda(t))}{\sigma^2\varphi(t)} \right)^2 dt \\ &\quad - \frac{1}{2} \int_0^T \left(\frac{b-r}{\sigma} \right)^2 \frac{\psi_\lambda(t)^2}{\varphi(t)} dt. \end{aligned}$$

Consequently,

$$\tilde{V}(\lambda) = \exp\left(-\frac{(b-r)^2}{\sigma^2}T\right) (\lambda - e^{rT}x)^2.$$

Proposition 3.2.1 (Conjugate relations). For every $\lambda, m \in \mathbb{R}$,

$$\tilde{V}(\lambda) = \inf_{q \in \mathbb{R}} \{V(q) + (q - \lambda)^2\},$$

and

$$V(m) = \sup_{\lambda \in \mathbb{R}} \{\tilde{V}(\lambda) - (m - \lambda)^2\}.$$

The maximizer in the second relation is

$$\lambda_m = \frac{m - \exp\left[\left(r - \frac{(b-r)^2}{\sigma^2}\right)T\right]x}{1 - \exp\left[-\frac{(b-r)^2}{\sigma^2}T\right]}.$$

The control $\hat{\alpha}_{\lambda_m}$ is optimal for $V(m)$.

Proof. For every admissible terminal wealth,

$$\mathbb{E}[(X_T - \lambda)^2] = \text{Var}(X_T) + (\mathbb{E}[X_T] - \lambda)^2.$$

Taking infima first over controls with a fixed mean and then over the mean gives the first conjugate relation. Convex duality gives the second.

Let λ_m maximize

$$\lambda \mapsto \tilde{V}(\lambda) - (m - \lambda)^2.$$

Then m minimizes

$$q \mapsto V(q) + (q - \lambda_m)^2.$$

Strict convexity implies

$$m = \mathbb{E}[\hat{X}_T^{\lambda_m}].$$

Using the conjugate relation with the correct sign,

$$\begin{aligned} V(m) &= \tilde{V}(\lambda_m) - (m - \lambda_m)^2 \\ &= \mathbb{E}[(\hat{X}_T^{\lambda_m} - \lambda_m)^2] - (\mathbb{E}[\hat{X}_T^{\lambda_m}] - \lambda_m)^2 \\ &= \text{Var}(\hat{X}_T^{\lambda_m}). \end{aligned}$$

Thus $\hat{\alpha}_{\lambda_m}$ is feasible and optimal for the mean–variance problem. □

3.2.2 Exponential-Utility Optimization with an Option Payoff

Consider a market with a risk-free asset normalized to $S^0 = 1$ and one risky asset

$$dS_t = S_t(b_t dt + \sigma_t dW_t).$$

The filtration is the augmented Brownian filtration. Assume that b and σ are bounded progressively measurable processes and that

$$\sigma_t \geq \varepsilon_0 > 0 \quad \text{a.s.}$$

An investor with initial capital x uses a progressively measurable strategy α , producing wealth

$$X_t^{x,\alpha} = x + \int_0^t \alpha_u(b_u du + \sigma_u dW_u).$$

Let $\mathcal{A}$ be a class of strategies satisfying

$$\int_0^T |\alpha_t|^2 dt < \infty \quad \text{a.s.}$$

and an admissibility condition strong enough to make the utility family below a supermartingale; for example, one may require the wealth to be bounded below by a deterministic constant. The

investor must deliver a bounded $\mathcal{F}_T$ -measurable claim ξ . With exponential utility

$$U(x) = -e^{-\eta x}, \quad \eta > 0,$$

the objective is

$$v(x) := \sup_{\alpha \in \mathcal{A}} \mathbb{E}[U(X_T^{x,\alpha} - \xi)].$$

The martingale optimality principle seeks processes J^α satisfying

- (i) $J_T^\alpha = U(X_T^{x,\alpha} - \xi)$;
- (ii) J_0^α is deterministic and independent of α ;
- (iii) each J^α is a supermartingale, and $J^{\hat{\alpha}}$ is a martingale for some admissible $\hat{\alpha}$.

When these conditions hold, $\hat{\alpha}$ is optimal and $v(x) = J_0^{\hat{\alpha}}$.

Let (Y, Z) solve the BSDE

$$Y_t = \xi + \int_t^T f(s, Z_s) ds - \int_t^T Z_s dW_s.$$

Assuming $\mathcal{F}_0$ is trivial, Y_0 is deterministic. Define the candidate family

$$J_t^\alpha := U(X_t^{x,\alpha} - Y_t).$$

Conditions (i) and (ii) are immediate, and the candidate value is $U(x - Y_0)$.

A direct Itô calculation gives the factorization

$$J_t^\alpha = M_t^\alpha C_t^\alpha,$$

where

$$\begin{aligned} M_t^\alpha &:= \exp[-\eta(x - Y_0)] \\ &\times \exp\left(-\int_0^t \eta(\alpha_u \sigma_u - Z_u) dW_u - \frac{1}{2} \int_0^t |\eta(\alpha_u \sigma_u - Z_u)|^2 du\right) \end{aligned}$$

is a positive local martingale and

$$C_t^\alpha := -\exp\left(\int_0^t \rho(u, \alpha_u, Z_u) du\right),$$

with

$$\rho(t, a, z) := \eta \left(\frac{\eta}{2} |a\sigma_t - z|^2 - ab_t - f(t, z) \right).$$

Completing the square,

$$\begin{aligned} \frac{1}{\eta} \rho(t, a, z) &= \frac{\eta}{2} \left| a\sigma_t - z - \frac{b_t}{\eta\sigma_t} \right|^2 \\ &\quad - z \frac{b_t}{\sigma_t} - \frac{1}{2\eta} \left| \frac{b_t}{\sigma_t} \right|^2 - f(t, z). \end{aligned}$$

Thus choose

$$f(t, z) = -z \frac{b_t}{\sigma_t} - \frac{1}{2\eta} \left| \frac{b_t}{\sigma_t} \right|^2$$

and

$$\hat{\alpha}_t = \frac{Z_t}{\sigma_t} + \frac{b_t}{\eta\sigma_t^2}.$$

Then $\rho(t, a, z) \geq 0$ for every a , and $\rho(t, \hat{\alpha}_t, Z_t) = 0$.

Theorem 3.2.2. Assume that the linear BSDE with terminal condition $Y_T = \xi$ and generator

$$f(t, z) = -z \frac{b_t}{\sigma_t} - \frac{1}{2\eta} \left| \frac{b_t}{\sigma_t} \right|^2$$

has its standard bounded solution and that the candidate strategy

$$\hat{\alpha}_t = \frac{Z_t}{\sigma_t} + \frac{b_t}{\eta\sigma_t^2}$$

belongs to the admissible class. Then

$$v(x) = U(x - Y_0) = -\exp[-\eta(x - Y_0)],$$

and $\hat{\alpha}$ is optimal.

Proof. For every admissible α , M^α is a positive local martingale. Since $\rho \geq 0$, the finite-variation factor C^α is nonincreasing, so the localized processes J^α are supermartingales. Under the admissibility condition and boundedness of Y , the process J^α is bounded below. Fatou's lemma therefore removes the localization and shows that J^α is a supermartingale.

For $\hat{\alpha}$, the drift factor vanishes and $C^{\hat{\alpha}} = -1$. Hence

$$\begin{aligned} J_t^{\hat{\alpha}} &= -M_t^{\hat{\alpha}} \\ &= -\exp[-\eta(x - Y_0)] \\ &\quad \times \exp\left(-\int_0^t \frac{b_u}{\sigma_u} dW_u - \frac{1}{2} \int_0^t \left| \frac{b_u}{\sigma_u} \right|^2 du\right). \end{aligned}$$

Because b/σ is bounded, Novikov's condition holds, so the stochastic exponential is a true martingale. Therefore $J^{\hat{\alpha}}$ is a martingale and the martingale optimality principle proves the result. $\square$

Remark. The optimal strategy decomposes as

$$\hat{\alpha}_t = \frac{Z_t}{\sigma_t} + \frac{b_t}{\eta\sigma_t^2}.$$

The first term is the liability-hedging component, while the second is the myopic investment component. The change of measure is generated by the market price of risk b_t/σ_t , not by Y itself. Under the corresponding martingale measure, the linear BSDE gives a conditional-expectation representation of the liability-adjusted certainty equivalent Y .

Chapter 4 Basic Theory of BSDEs

4.1 Introduction to BSDEs

4.1.1 Existence and uniqueness

Let $B = (B_t)_{0 \leq t \leq T}$ be a d -dimensional Brownian motion on a filtered probability space $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$, where $\mathbb{F} = (\mathcal{F}_t)_{0 \leq t \leq T}$ is the usual augmentation of the natural filtration of B , and let $T < \infty$.

We denote by $\mathbb{S}^2(0, T)$ the space of real-valued, continuous, $\mathbb{F}$ -adapted processes Y such that

$$\mathbb{E} \left[\sup_{0 \leq t \leq T} |Y_t|^2 \right] < \infty,$$

and by $\mathbb{H}^2(0, T; \mathbb{R}^d)$ the space of progressively measurable $\mathbb{R}^d$ -valued processes Z such that

$$\mathbb{E} \left[\int_0^T |Z_t|^2 dt \right] < \infty.$$

We are given a terminal condition ξ and a generator f satisfying:

- (A) $\xi \in L^2(\Omega, \mathcal{F}_T, \mathbb{P})$;
- (B) for every $(y, z) \in \mathbb{R} \times \mathbb{R}^d$, the process $(t, \omega) \mapsto f(\omega, t, y, z)$ is progressively measurable,

$$\mathbb{E} \left[\int_0^T |f(t, 0, 0)|^2 dt \right] < \infty,$$

and there is a constant L_f such that, for all $y_1, y_2 \in \mathbb{R}$ and $z_1, z_2 \in \mathbb{R}^d$,

$$|f(t, y_1, z_1) - f(t, y_2, z_2)| \leq L_f (|y_1 - y_2| + |z_1 - z_2|), \quad dt \otimes d\mathbb{P}\text{-a.e.}$$

Consider the one-dimensional BSDE

$$-dY_t = f(t, Y_t, Z_t) dt - Z_t \cdot dB_t, \quad Y_T = \xi.$$

Definition 4.1.1. A solution is a pair $(Y, Z) \in \mathbb{S}^2(0, T) \times \mathbb{H}^2(0, T; \mathbb{R}^d)$ such that

$$Y_t = \xi + \int_t^T f(s, Y_s, Z_s) ds - \int_t^T Z_s \cdot dB_s, \quad 0 \leq t \leq T,$$

almost surely.

Theorem 4.1.2 (Pardoux–Peng). Under assumptions (A) and (B), the BSDE admits a unique solution in $\mathbb{S}^2(0, T) \times \mathbb{H}^2(0, T; \mathbb{R}^d)$.

Proof. For progressively measurable $(U, V) \in \mathbb{H}^2(0, T; \mathbb{R}) \times \mathbb{H}^2(0, T; \mathbb{R}^d)$, set

$$M_t := \mathbb{E} \left[\xi + \int_0^T f(s, U_s, V_s) \, ds \mid \mathcal{F}_t \right].$$

The random variable inside the conditional expectation is square integrable. By the martingale representation theorem, there is a unique $Z \in \mathbb{H}^2(0, T; \mathbb{R}^d)$ such that

$$M_t = M_0 + \int_0^t Z_s \cdot dB_s.$$

Define

$$Y_t := M_t - \int_0^t f(s, U_s, V_s) \, ds.$$

Then

$$Y_t = \xi + \int_t^T f(s, U_s, V_s) \, ds - \int_t^T Z_s \cdot dB_s.$$

Doob's inequality, Cauchy–Schwarz, and the Lipschitz condition imply $Y \in \mathbb{S}^2(0, T)$. Thus this construction defines a map $\Phi(U, V) = (Y, Z)$.

Let $\Phi(U, V) = (Y, Z)$ and $\Phi(U', V') = (Y', Z')$, and write bars for differences. Since $\bar{Y}_T = 0$, Itô's formula gives, for every $t \in [0, T]$,

$$\begin{aligned} e^{\beta t} |\bar{Y}_t|^2 + \int_t^T e^{\beta s} (\beta |\bar{Y}_s|^2 + |\bar{Z}_s|^2) \, ds \\ = 2 \int_t^T e^{\beta s} \bar{Y}_s \bar{f}_s \, ds - 2 \int_t^T e^{\beta s} \bar{Y}_s \bar{Z}_s \cdot dB_s, \end{aligned}$$

where $\bar{f}_s = f(s, U_s, V_s) - f(s, U'_s, V'_s)$. After localization and expectation, Young's inequality yields

$$\begin{aligned} \mathbb{E} \left[\int_0^T e^{\beta s} (|\bar{Y}_s|^2 + |\bar{Z}_s|^2) \, ds \right] \\ \leq \frac{1}{2} \mathbb{E} \left[\int_0^T e^{\beta s} (|\bar{U}_s|^2 + |\bar{V}_s|^2) \, ds \right] \end{aligned}$$

when, for example, $\beta = 2 + 8L_f^2$. Indeed, the remaining factor on the right is at most $4L_f^2/\beta < 1/2$. Hence Φ is a contraction on the Banach space $\mathbb{H}^2(0, T; \mathbb{R}) \times \mathbb{H}^2(0, T; \mathbb{R}^d)$ endowed with the corresponding weighted norm. Its fixed point solves the BSDE, and the preceding estimate also proves uniqueness. The construction already shows that the fixed-point process Y belongs to $\mathbb{S}^2(0, T)$. $\square$

4.1.2 Explicit solution of linear BSDEs

Consider

$$-dY_t = (A_t Y_t + B_t \cdot Z_t + C_t) \, dt - Z_t \cdot dB_t, \quad Y_T = \xi,$$

where A and B are bounded progressively measurable processes with values in $\mathbb{R}$ and $\mathbb{R}^d$, respectively, and $C \in \mathbb{H}^2(0, T; \mathbb{R})$.

Proposition 4.1.3. Let Γ solve

$$d\Gamma_t = \Gamma_t(A_t dt + B_t \cdot dB_t), \quad \Gamma_0 = 1.$$

Then the unique solution of the linear BSDE satisfies

$$\Gamma_t Y_t = \mathbb{E} \left[\Gamma_T \zeta + \int_t^T \Gamma_s C_s ds \mid \mathcal{F}_t \right].$$

Proof. Itô's product formula gives

$$d(\Gamma_t Y_t) = -\Gamma_t C_t dt + \Gamma_t(Z_t + Y_t B_t) \cdot dB_t.$$

Consequently,

$$\Gamma_t Y_t + \int_0^t \Gamma_s C_s ds = Y_0 + \int_0^t \Gamma_s(Z_s + Y_s B_s) \cdot dB_s.$$

The stochastic integral is a uniformly integrable martingale: boundedness of A and B gives finite moments of $\sup_{t \leq T} |\Gamma_t|$, while $(Y, Z) \in \mathbb{S}^2 \times \mathbb{H}^2$. Taking conditional expectations at time t proves the formula. The process Z is then identified by martingale representation. $\square$

4.1.3 Comparison principles

Theorem 4.1.4 (Comparison). For $i = 1, 2$, let (ζ^i, f^i) satisfy (A)–(B), and let (Y^i, Z^i) be the corresponding solutions. Assume

$$\zeta^1 \leq \zeta^2 \quad \text{a.s.}$$

and

$$f^1(t, Y_t^1, Z_t^1) \leq f^2(t, Y_t^1, Z_t^1), \quad dt \otimes d\mathbb{P}\text{-a.e.}$$

Then $Y_t^1 \leq Y_t^2$ for every $t \in [0, T]$, almost surely. If $\mathcal{F}_0$ is trivial and either $\mathbb{P}(\zeta^1 < \zeta^2) > 0$ or the generator inequality is strict on a set of positive $dt \otimes d\mathbb{P}$ measure, then $Y_0^1 < Y_0^2$.

Proof. Set $\bar{Y} = Y^2 - Y^1$ and $\bar{Z} = Z^2 - Z^1$. For clarity, first take a one-dimensional Brownian motion. Define bounded divided differences Δ^y and Δ^z for f^2 and set

$$\bar{f}_t := f^2(t, Y_t^1, Z_t^1) - f^1(t, Y_t^1, Z_t^1) \geq 0.$$

Then

$$-d\bar{Y}_t = (\Delta_t^y \bar{Y}_t + \Delta_t^z \bar{Z}_t + \bar{f}_t) dt - \bar{Z}_t dB_t, \quad \bar{Y}_T = \zeta^2 - \zeta^1.$$

The linear representation above gives

$$\Gamma_t \bar{Y}_t = \mathbb{E} \left[\Gamma_T (\zeta^2 - \zeta^1) + \int_t^T \Gamma_s \bar{f}_s ds \mid \mathcal{F}_t \right],$$

where $\Gamma > 0$. The result follows. For $d > 1$, replace the scalar z -divided difference by the vector

$$\Delta_t^z := \mathbf{1}_{\{\bar{Z}_t \neq 0\}} \frac{f^2(t, Y_t^1, Z_t^2) - f^2(t, Y_t^1, Z_t^1)}{|\bar{Z}_t|^2} \bar{Z}_t,$$

so that the same linear equation contains $\Delta_t^z \cdot \bar{Z}_t$. The strict assertion follows from positivity of Γ and triviality of $\mathcal{F}_0$. $\square$

Remark. Only the generator used in the linearization must be Lipschitz. Thus the proof still works when f^2 is Lipschitz and f^1 merely satisfies the integrability conditions needed for its BSDE.

Corollary 4.1.5. If $\xi \geq 0$ almost surely and $f(t, 0, 0) \geq 0$ $dt \otimes d\mathbb{P}$ -a.e., then

$$Y_t \geq 0, \quad 0 \leq t \leq T, \quad \text{a.s.}$$

If $\mathcal{F}_0$ is trivial and either $\mathbb{P}(\xi > 0) > 0$ or $f(t, 0, 0) > 0$ on a set of positive $dt \otimes d\mathbb{P}$ measure, then $Y_0 > 0$.

4.1.4 Stability estimates

We now allow a multidimensional solution. Consider

$$Y_t = \xi + \int_t^T f_s(Y_s, Z_s) ds - \int_t^T Z_s dB_s, \quad 0 \leq t \leq T,$$

where Y takes values in $\mathbb{R}^{d_2}$ and Z in $\mathbb{R}^{d_2 \times d}$. Assume that f is progressively measurable in (t, ω) , Borel measurable in (y, z) , uniformly Lipschitz in (y, z) with constant L , and

$$\xi \in L^2(\mathcal{F}_T; \mathbb{R}^{d_2}), \quad I_0^2 := \mathbb{E} \left[|\xi|^2 + \left(\int_0^T |f_t(0, 0)| dt \right)^2 \right] < \infty.$$

Write $Y_t^* := \sup_{0 \leq s \leq t} |Y_s|$.

Theorem 4.1.6 (A priori estimate). There is a constant C , depending only on T, L, d , and d_2 , such that

$$\mathbb{E} \left[|Y_T^*|^2 + \int_0^T |Z_t|^2 dt \right] \leq C I_0^2.$$

Proof. The linear-growth bound $|f_t(y, z)| \leq |f_t(0, 0)| + L(|y| + |z|)$, the BSDE, and the BDG inequality give

$$\mathbb{E} |Y_T^*|^2 \leq C \mathbb{E} \left[|\xi|^2 + \left(\int_0^T |f_t(0, 0)| dt \right)^2 + \int_0^T (|Y_t|^2 + |Z_t|^2) dt \right].$$

Itô's formula for $|Y_t|^2$, followed by expectation, Young's inequality, and backward Gronwall, yields, for every $\varepsilon > 0$,

$$\sup_{0 \leq t \leq T} \mathbb{E} |Y_t|^2 + \mathbb{E} \left[\int_0^T |Z_t|^2 dt \right] \leq \varepsilon \mathbb{E} |Y_T^*|^2 + C_\varepsilon I_0^2.$$

Substitution into the first inequality and absorption of the $\mathbb{E} |Y_T^*|^2$ term prove the estimate. $\square$

Remark. The preceding a priori estimate uses only the linear-growth bound on f . Lipschitz continuity is needed for uniqueness and for the stability results below.

Corollary 4.1.7 (Stability). For $i = 1, 2$, let (Y^i, Z^i) solve the BSDE with data (ξ_i, f^i) , where both generators have the same Lipschitz bound. With

$$\Delta Y := Y^1 - Y^2, \quad \Delta Z := Z^1 - Z^2, \quad \Delta \xi := \xi_1 - \xi_2,$$

$$\Delta f_t(y, z) := f_t^1(y, z) - f_t^2(y, z),$$

there is a constant C such that

$$\mathbb{E} \left[|(\Delta Y)_T^*|^2 + \int_0^T |\Delta Z_t|^2 dt \right] \leq C \mathbb{E} \left[|\Delta \xi|^2 + \left(\int_0^T |\Delta f_t(Y_t^1, Z_t^1)| dt \right)^2 \right].$$

Theorem 4.1.8 (Convergence of solutions). Let (ξ_n, f^n) and (ξ, f) satisfy the preceding assumptions with a common Lipschitz constant, and let (Y^n, Z^n) and (Y, Z) be the corresponding solutions. Assume

$$\mathbb{E} \left[|\xi_n - \xi|^2 + \left(\int_0^T |f_t^n(0, 0) - f_t(0, 0)| dt \right)^2 \right] \rightarrow 0$$

and, for every fixed (y, z) , $f^n(\cdot, y, z) \rightarrow f(\cdot, y, z)$ in $dt \otimes d\mathbb{P}$ measure. Then

$$\mathbb{E} \left[|(Y^n - Y)_T^*|^2 + \int_0^T |Z_t^n - Z_t|^2 dt \right] \rightarrow 0.$$

Proof. Apply the stability estimate with the reference solution (Y, Z) . The common Lipschitz constant gives

$$|(f_t^n - f_t)(Y_t, Z_t) - (f_t^n - f_t)(0, 0)| \leq 2L(|Y_t| + |Z_t|).$$

Approximation of (Y, Z) by simple processes, together with convergence in measure for fixed (y, z) , shows convergence in measure of the left side to zero. Its time integral squared is uniformly integrable by the right-hand bound and the L^2 estimates for (Y, Z) . The generalized dominated convergence theorem and the stability estimate complete the proof. $\square$

Theorem 4.1.9 (L^p estimate). Let $p \geq 2$ and assume

$$I_p^p := \mathbb{E} \left[|\xi|^p + \left(\int_0^T |f_t(0, 0)| dt \right)^p \right] < \infty.$$

Then the unique solution satisfies

$$\mathbb{E} \left[|Y_T^*|^p + \left(\int_0^T |Z_t|^2 dt \right)^{p/2} \right] \leq C_p I_p^p.$$

Proof. For bounded data, Itô's formula for $|Y|^p$, the BDG inequality, Young's inequality, and the

same coupled-estimate argument as in the L^2 case give

$$\mathbb{E}|Y_T^*|^p + \mathbb{E} \left[\int_0^T |Y_t|^{p-2} |Z_t|^2 dt \right] \leq C_p I_p^p.$$

Applying Itô's formula to $|Y|^2$ and then raising the resulting estimate to the power $p/2$ yields

$$\begin{aligned} \mathbb{E} \left[\left(\int_0^T |Z_t|^2 dt \right)^{p/2} \right] &\leq C_p I_p^p + C_p \mathbb{E}|Y_T^*|^p \\ &\quad + \frac{1}{2} \mathbb{E} \left[\left(\int_0^T |Z_t|^2 dt \right)^{p/2} \right]. \end{aligned}$$

Absorption gives the desired Z estimate. General data follow by truncating ξ and f , applying the bounded-data estimate uniformly, using L^2 stability, and then applying Fatou's lemma. $\square$

4.2 Decoupled FBSDE systems

Let B be d -dimensional and, for $0 \leq t \leq s \leq T$, write $B_{t,s} := B_s - B_t$ and

$$\mathcal{F}_s^t := \sigma(B_{t,r} : t \leq r \leq s) \vee \mathcal{N}, \quad \mathbb{F}^t = (\mathcal{F}_s^t)_{t \leq s \leq T},$$

where $\mathcal{N}$ denotes the null sets. Then $\mathbb{F}^t$ is independent of $\mathcal{F}_t$.

Consider the Markovian decoupled FBSDE

$$\begin{cases} X_s^{t,x} = x + \int_t^s b(r, X_r^{t,x}) dr + \int_t^s \sigma(r, X_r^{t,x}) dB_r, \\ Y_s^{t,x} = g(X_T^{t,x}) + \int_s^T f(r, X_r^{t,x}, Y_r^{t,x}, Z_r^{t,x}) dr - \int_s^T Z_r^{t,x} dB_r. \end{cases}$$

Here X, Y, Z take values in $\mathbb{R}^{d_1}, \mathbb{R}^{d_2}$, and $\mathbb{R}^{d_2 \times d}$, respectively. Assume that b, σ, f, g are deterministic; $b(\cdot, 0), \sigma(\cdot, 0), f(\cdot, 0, 0, 0)$, and $g(0)$ are bounded; the coefficients are uniformly Lipschitz in their spatial variables; and b, σ, f are uniformly $1/2$ -Hölder continuous in time. The time regularity is not needed for well-posedness, but it is used below.

For $\eta \in L^2(\mathcal{F}_t; \mathbb{R}^{d_1})$, let $(\mathcal{X}^{t,\eta}, \mathcal{Y}^{t,\eta}, \mathcal{Z}^{t,\eta})$ denote the solution obtained by replacing x with η .

4.2.1 The Markov property

Theorem 4.2.1. Fix $t \in [0, T]$.

- (i) The family $(x, s, \omega) \mapsto (Y_s^{t,x}(\omega), Z_s^{t,x}(\omega))$ admits a jointly measurable version that is $\mathbb{F}^t$ -progressively measurable in (s, ω) for every x . In particular, for fixed x , it is independent of $\mathcal{F}_t$.
- (ii) For every $\eta \in L^2(\mathcal{F}_t; \mathbb{R}^{d_1})$,

$$\mathcal{Y}_s^{t,\eta}(\omega) = Y_s^{t,\eta(\omega)}(\omega), \quad \mathcal{Z}_s^{t,\eta}(\omega) = Z_s^{t,\eta(\omega)}(\omega),$$

for $ds \otimes d\mathbb{P}$ -a.e. (s, ω) , with the equality for Y holding up to indistinguishability.

(iii) The function

$$u(t, x) := Y_t^{t, x}$$

is deterministic and Borel measurable, and the flow property gives

$$Y_s^{t, x} = u(s, X_s^{t, x}), \quad t \leq s \leq T.$$

Proof. Picard iteration for the forward SDE and the BSDE may be carried out using only the increments $(B_r - B_t)_{r \geq t}$, which gives an $\mathbb{F}^t$ -adapted measurable version and proves (i). First check (ii) for simple $\mathcal{F}_t$ -measurable random variables η by pasting the solutions on the atoms of the simple variable. The general case follows from the stability estimates. Since $Y_t^{t, x}$ is both $\mathcal{F}_t$ -measurable and independent of $\mathcal{F}_t$, it is constant almost surely, proving that u is deterministic. Finally, uniqueness applied after time s yields the flow identity and hence $Y_s^{t, x} = u(s, X_s^{t, x})$. $\square$

Remark. Under additional regularity, Z also has the Markovian representation $Z_s^{t, x} = v(s, X_s^{t, x})$ for $ds \otimes d\mathbb{P}$ -a.e. (s, ω) . Without such a choice of version, it is safer to state the Markov property through the forward state and the decoupling field u .

4.2.2 Regularity of solutions

Theorem 4.2.2 (State stability). There is a constant C such that, for all t, x, x_1, x_2 ,

$$\mathbb{E} \left[\sup_{t \leq s \leq T} |X_s^{t, x}|^2 + \sup_{t \leq s \leq T} |Y_s^{t, x}|^2 + \int_t^T |Z_s^{t, x}|^2 ds \right] \leq C(1 + |x|^2),$$

and

$$\mathbb{E} \left[\sup_{t \leq s \leq T} |X_s^{t, x_1} - X_s^{t, x_2}|^2 + \sup_{t \leq s \leq T} |Y_s^{t, x_1} - Y_s^{t, x_2}|^2 + \int_t^T |Z_s^{t, x_1} - Z_s^{t, x_2}|^2 ds \right] \leq C|x_1 - x_2|^2.$$

Consequently,

$$|u(t, x)| \leq C(1 + |x|), \quad |u(t, x_1) - u(t, x_2)| \leq C|x_1 - x_2|.$$

Proof. Apply the standard SDE estimates to the forward equation and then the BSDE a priori and stability estimates to the backward equation. Since $u(t, x) = Y_t^{t, x}$ is deterministic, the displayed bounds for u follow by evaluating the corresponding L^2 estimates at time t . $\square$

Theorem 4.2.3 (Time stability). There is a constant C such that

$$|Z_t| \leq C|\sigma(t, X_t)| \leq C(1 + |X_t|), \quad dt \otimes d\mathbb{P}\text{-a.e.}$$

Moreover, for $t_1 < t_2$,

$$\mathbb{E} \left[\sup_{t_1 \leq s \leq t_2} |X_s - X_{t_1}|^2 + \sup_{t_1 \leq s \leq t_2} |Y_s - Y_{t_1}|^2 \right] \leq C(1 + |x|^2)(t_2 - t_1),$$

and

$$|u(t_1, x) - u(t_2, x)| \leq C(1 + |x|)\sqrt{t_2 - t_1}.$$

Proof. For smooth coefficients, differentiability of the stochastic flow gives $Z_t = D_x u(t, X_t) \sigma(t, X_t)$, and the state-stability estimate bounds $D_x u$. Approximation by smooth coefficients and L^2 stability yield the general result. The increment estimates then follow from the SDE and BSDE equations, BDG, and the bound on Z . Finally, use the flow property at t_2 :

$$u(t_1, x) - u(t_2, x) = \mathbb{E} \left[Y_{t_1}^{t_1, x} - Y_{t_2}^{t_1, x} + u(t_2, X_{t_2}^{t_1, x}) - u(t_2, x) \right],$$

and apply the preceding estimates and the spatial Lipschitz bound for u . $\square$

Lemma 4.2.4 (Derivative representation). Assume, in addition, that b and σ are continuously differentiable in x , that f is continuously differentiable in (x, y, z) , and that g is continuously differentiable in x , with all first derivatives bounded. Then $u(t, \cdot)$ is continuously differentiable and

$$D_x u(t, X_t) = \nabla Y_t (\nabla X_t)^{-1}, \quad Z_t = \nabla Y_t (\nabla X_t)^{-1} \sigma(t, X_t),$$

where the variational processes solve

$$\begin{aligned} \nabla X_t &= I_{d_1} + \int_0^t D_x b(s, X_s) \nabla X_s \, ds + \sum_{k=1}^d \int_0^t D_x \sigma^k(s, X_s) \nabla X_s \, dB_s^k, \\ \nabla Y_t &= D_x g(X_T) \nabla X_T \\ &\quad + \int_t^T \left[D_x f \nabla X_s + D_y f \nabla Y_s + \sum_{k=1}^d D_{z_k} f \nabla Z_s^k \right] (s, X_s, Y_s, Z_s) \, ds \\ &\quad - \sum_{k=1}^d \int_t^T \nabla Z_s^k \, dB_s^k. \end{aligned}$$

Proof. Difference quotients of the forward and backward equations satisfy linear FBSDEs whose coefficients are averaged first derivatives. The state-stability estimates show that these difference quotients converge in the natural L^2 spaces to the displayed variational solution. In particular,

$$D_x u(0, x) = \nabla Y_0,$$

and the same argument after an arbitrary starting time proves continuous differentiability of $u(t, \cdot)$. The flow identity gives

$$\nabla Y_t = D_x u(t, X_t) \nabla X_t.$$

The Jacobian ∇X_t is invertible for the stochastic flow under the stated assumptions. Applying Itô's formula to $u(t, X_t)$ in the smooth case gives the formula for Z ; approximation gives the general case. $\square$

4.3 The nonlinear Feynman–Kac formula

Let

$$\mathcal{L}\varphi(t, x) := b(x) \cdot D_x \varphi(t, x) + \frac{1}{2} \operatorname{tr} \left(\sigma(x) \sigma(x)^\top D_{xx}^2 \varphi(t, x) \right).$$

The linear Feynman–Kac formula represents the solution of

$$\begin{cases} -\partial_t v - \mathcal{L}v - f(t, x) = 0, & (t, x) \in [0, T) \times \mathbb{R}^n, \\ v(T, x) = g(x), & x \in \mathbb{R}^n, \end{cases}$$

as

$$v(t, x) = \mathbb{E} \left[g(X_T^{t,x}) + \int_t^T f(s, X_s^{t,x}) \, ds \right].$$

We now consider the semilinear equation

$$\begin{cases} -\partial_t v - \mathcal{L}v - f(t, x, v, \sigma(x)^\top D_x v) = 0, & (t, x) \in [0, T) \times \mathbb{R}^n, \\ v(T, x) = g(x), & x \in \mathbb{R}^n. \end{cases}$$

Assume that b and σ are Lipschitz, that f is continuous in (t, x) , uniformly Lipschitz in (y, z) , and of linear growth, and that g is continuous with linear growth. Let

$$\begin{aligned} dX_s^{t,x} &= b(X_s^{t,x}) \, ds + \sigma(X_s^{t,x}) \, dB_s, & X_t^{t,x} &= x, \\ -dY_s^{t,x} &= f(s, X_s^{t,x}, Y_s^{t,x}, Z_s^{t,x}) \, ds - Z_s^{t,x} \, dB_s, \\ Y_T^{t,x} &= g(X_T^{t,x}), \end{aligned}$$

and define $v(t, x) := Y_t^{t,x}$.

Proposition 4.3.1 (Classical nonlinear Feynman–Kac formula). Suppose $v \in C^{1,2}([0, T) \times \mathbb{R}^n) \cap C([0, T] \times \mathbb{R}^n)$ is a classical solution of the semilinear PDE, has linear growth, and satisfies

$$|D_x v(t, x)| \leq C(1 + |x|^q)$$

for some constants $C, q > 0$. Then

$$Y_t = v(t, X_t), \quad Z_t = \sigma(X_t)^\top D_x v(t, X_t)$$

is the unique solution of the associated BSDE.

Proof. Apply Itô's formula to $v(t, X_t)$. The PDE identifies the drift and the stochastic term. The growth assumptions and the moment estimates for X imply $(Y, Z) \in \mathbb{S}^2 \times \mathbb{H}^2$, so uniqueness of the BSDE completes the proof. $\square$

Proposition 4.3.2 (Viscosity nonlinear Feynman–Kac formula). The function $v(t, x) = Y_t^{t,x}$ is continuous on $[0, T] \times \mathbb{R}^n$ and is a viscosity solution of the semilinear PDE.

Proof. Continuity follows directly from the state- and time-stability estimates proved above.

We verify the subsolution property; the supersolution argument is analogous. Let $\varphi \in C^{1,2}$ and suppose that $v - \varphi$ has a local maximum equal to zero at (t, x) . Assume, for contradiction, that

$$-\partial_t \varphi(t, x) - \mathcal{L}\varphi(t, x) - f(t, x, v(t, x), \sigma(x)^\top D_x \varphi(t, x)) > 0.$$

By continuity and the equality $v(t, x) = \varphi(t, x)$, there are $h, \varepsilon, \delta > 0$ such that, whenever $t \leq s \leq t + h$ and $|y - x| \leq \varepsilon$,

$$v(s, y) \leq \varphi(s, y)$$

and

$$\begin{aligned} & f(s, y, \varphi(s, y), \sigma(y)^\top D_x \varphi(s, y)) + \partial_t \varphi(s, y) + \mathcal{L}\varphi(s, y) \\ & \leq -\delta. \end{aligned}$$

Let

$$\tau := \inf\{s \geq t : |X_s^{t,x} - x| \geq \varepsilon\} \wedge (t + h).$$

By the flow property, the restriction of $(Y^{t,x}, Z^{t,x})$ to $[t, \tau]$ is the BSDE with terminal value $v(\tau, X_\tau^{t,x})$. Define

$$\begin{aligned} \bar{Y}_s &:= Y_s^{t,x} - \varphi(s, X_s^{t,x}), \\ \bar{Z}_s &:= Z_s^{t,x} - \sigma(X_s^{t,x})^\top D_x \varphi(s, X_s^{t,x}), \quad t \leq s \leq \tau. \end{aligned}$$

Then $\bar{Y}_\tau \leq 0$ and

$$\bar{Y}_s = \bar{Y}_\tau + \int_s^\tau F(r, \bar{Y}_r, \bar{Z}_r) dr - \int_s^\tau \bar{Z}_r dB_r,$$

where, writing $X_r = X_r^{t,x}$ and $z_r^\varphi = \sigma(X_r)^\top D_x \varphi(r, X_r)$,

$$\begin{aligned} F(r, y, z) &:= f(r, X_r, y + \varphi(r, X_r), z + z_r^\varphi) \\ &\quad + \partial_t \varphi(r, X_r) + \mathcal{L}\varphi(r, X_r). \end{aligned}$$

The generator F is Lipschitz in (y, z) and $F(r, 0, 0) \leq -\delta$ on $[t, \tau]$. Let $(\tilde{Y}, \tilde{Z})$ solve the same BSDE with terminal value zero. Comparison gives $\bar{Y}_t \leq \tilde{Y}_t$. Linearizing F around $(0, 0)$ and using the linear representation formula yields

$$\tilde{Y}_t = \mathbb{E}_t^{\mathbb{Q}} \left[\int_t^\tau \exp \left(\int_t^s a_r dr \right) F(s, 0, 0) ds \right] < 0,$$

for an equivalent measure $\mathbb{Q}$ and a bounded process a . Consequently,

$$0 = v(t, x) - \varphi(t, x) = \bar{Y}_t < 0,$$

a contradiction. Thus the viscosity subsolution inequality holds. $\square$

Chapter 5 BSDEs with Quadratic Growth in Z

5.1 A priori estimates and BMO analysis

We study scalar BSDEs of the form

$$Y_t = \xi + \int_t^T f_s(Y_s, Z_s) ds - \int_t^T Z_s dB_s, \quad 0 \leq t \leq T,$$

where B is d -dimensional and $\mathbb{F} = \mathbb{F}^B$ is its usual augmented filtration. For every (y, z) , assume that $(t, \omega) \mapsto f_t(\omega, y, z)$ is progressively measurable. Throughout this chapter, impose the following conditions, with a constant C_0 independent of (t, ω, y, z) :

(Q1)

$$|f_t(y, z)| \leq C_0(1 + |y| + |z|^2).$$

(Q2) For all (y_i, z_i) ,

$$\begin{aligned} |f_t(y_1, z_1) - f_t(y_2, z_2)| &\leq C_0(|y_1 - y_2| \\ &\quad + (1 + |y_1| + |y_2| + |z_1| + |z_2|)|z_1 - z_2|). \end{aligned}$$

(Q3) $\xi \in L^\infty(\mathcal{F}_T)$.

All inequalities involving the random generator are understood $dt \otimes d\mathbb{P}$ -almost everywhere.

Remark. If f is differentiable in (y, z) , condition (Q2) follows from

$$|\partial_y f_t(y, z)| \leq C_0, \quad |\partial_z f_t(y, z)| \leq C_0(1 + |y| + |z|),$$

and conversely these derivative bounds follow from (Q2) at points of differentiability.

Theorem 5.1.1 (A priori estimate). Let $(Y, Z) \in \mathbb{L}^\infty(\mathbb{F}; \mathbb{R}) \times \mathbb{H}^2(0, T; \mathbb{R}^d)$ solve the quadratic BSDE. Then

$$|Y_t| \leq C \quad \text{and} \quad \mathbb{E}_\tau \left[\int_\tau^T |Z_s|^2 ds \right] \leq C$$

for every stopping time $\tau \leq T$. The constant depends only on T, C_0 , and $\|\xi\|_\infty$.

Proof. Fix $t_0 \in [0, T]$ and define

$$\tau := \inf\{t \geq t_0 : |Y_t| \leq 1\} \wedge T.$$

Choose a smooth function $\psi \geq 0$ such that $\psi(y) = |y|$ for $|y| \geq 1$ and $|y| \leq \psi(y) + 1 \leq |y| + 2$ for

all y . Let

$$\lambda := 2(1 + C_0), \quad \varphi(t, y) := \lambda e^{\lambda t}(\psi(y) + 1), \quad \tilde{Y}_t := e^{\varphi(t, Y_t)}.$$

With $\tilde{Z}_t := \partial_y \varphi(t, Y_t) \tilde{Y}_t Z_t$, Itô's formula gives

$$d\tilde{Y}_t = \tilde{Y}_t \left(\partial_t \varphi - \partial_y \varphi f_t(Y_t, Z_t) + \frac{1}{2} \partial_{yy} \varphi |Z_t|^2 + \frac{1}{2} |\partial_y \varphi|^2 |Z_t|^2 \right) dt + \tilde{Z}_t dB_t.$$

On $[t_0, \tau)$, $|Y_t| > 1$, so $\psi(Y_t) = |Y_t|$ and $\partial_{yy} \varphi(t, Y_t) = 0$. The choice of λ and (Q1) make the drift nonnegative. Thus the stopped process $\tilde{Y}_{\cdot \wedge \tau}$ is a submartingale. Optional sampling gives

$$\begin{aligned} \exp\left(\lambda e^{\lambda t_0} |Y_{t_0}|\right) &\leq \tilde{Y}_{t_0} \leq \mathbb{E}_{t_0}[\tilde{Y}_\tau] \\ &\leq \exp\left(\lambda e^{\lambda T} (\|\xi\|_\infty + 2)\right). \end{aligned}$$

Hence

$$|Y_{t_0}| \leq e^{\lambda(T-t_0)} (\|\xi\|_\infty + 2),$$

which proves the uniform estimate for Y .

Apply Itô's formula to $e^{\lambda Y_t}$. Since $\lambda^2/2 - \lambda C_0 = \lambda$, condition (Q1) gives

$$de^{\lambda Y_t} \geq \lambda e^{\lambda Y_t} |Z_t|^2 dt - \lambda C_0 e^{\lambda Y_t} (1 + |Y_t|) dt + \lambda e^{\lambda Y_t} Z_t dB_t.$$

Stopping first, taking conditional expectations, and then removing the localization yields, for every stopping time τ ,

$$e^{\lambda Y_\tau} + \mathbb{E}_\tau \left[\int_\tau^T \lambda e^{\lambda Y_s} |Z_s|^2 ds \right] \leq \mathbb{E}_\tau \left[e^{\lambda \xi} + \int_\tau^T \lambda C_0 e^{\lambda Y_s} (1 + |Y_s|) ds \right].$$

The already established bound on Y now implies the conditional estimate for Z . $\square$

Definition 5.1.2. A continuous martingale M with $M_0 = 0$ is a BMO martingale if

$$\|M\|_{\text{BMO}}^2 := \sup_{\tau \in \mathcal{T}_{0,T}} \left\| \mathbb{E}_\tau[(M_T - M_\tau)^2] \right\|_\infty < \infty.$$

For $M = \int_0^\cdot \theta_s dB_s$, this is equivalent to

$$\sup_{\tau \in \mathcal{T}_{0,T}} \left\| \mathbb{E}_\tau \left[\int_\tau^T |\theta_s|^2 ds \right] \right\|_\infty < \infty.$$

The a priori estimate therefore shows that $\int_0^\cdot Z_s dB_s$ is a BMO martingale.

Theorem 5.1.3 (John–Nirenberg and reverse Hölder estimates). Let θ be progressively measurable and assume

$$\sup_{\tau \in \mathcal{T}_{0,T}} \left\| \mathbb{E}_\tau \left[\int_\tau^T |\theta_s|^2 ds \right] \right\|_\infty \leq C_0.$$

Then, for every stopping time τ and every $0 < \varepsilon < C_0^{-1}$,

$$\mathbb{E}_\tau \left[\exp \left(\varepsilon \int_\tau^T |\theta_s|^2 ds \right) \right] \leq \frac{1}{1 - C_0 \varepsilon}.$$

Moreover, the stochastic exponential

$$\mathcal{E}_t(\theta) := \exp \left(\int_0^t \theta_s dB_s - \frac{1}{2} \int_0^t |\theta_s|^2 ds \right)$$

is a uniformly integrable martingale and satisfies a reverse Hölder inequality: there are $p > 1$ and $C_p < \infty$, depending only on the BMO bound, such that

$$\sup_{\tau \in \mathcal{T}_{0,T}} \mathbb{E}_\tau \left[\left(\frac{\mathcal{E}_T(\theta)}{\mathcal{E}_\tau(\theta)} \right)^p \right] \leq C_p.$$

In particular, with $\mathcal{E}_T^*(\theta) := \sup_{0 \leq t \leq T} \mathcal{E}_t(\theta)$,

$$\mathbb{E}[(\mathcal{E}_T^*(\theta))^p] < \infty$$

for a possibly smaller exponent $p > 1$.

5.2 Comparison and stability

Theorem 5.2.1 (Comparison). For $i = 1, 2$, let (ξ_i, f^i) satisfy (Q1)–(Q3) with common constants, and let $(Y^i, Z^i) \in \mathbb{L}^\infty \times \mathbb{H}^2$ be the corresponding solutions. If

$$\xi_1 \leq \xi_2 \quad \text{a.s.} \quad \text{and} \quad f_t^1(y, z) \leq f_t^2(y, z)$$

for all (y, z) , $dt \otimes d\mathbb{P}$ -a.e., then $Y_t^1 \leq Y_t^2$ for every t , almost surely.

Proof. Set $\Delta Y = Y^1 - Y^2$, $\Delta Z = Z^1 - Z^2$, and

$$\delta f_t := f_t^1(Y_t^1, Z_t^1) - f_t^2(Y_t^1, Z_t^1) \leq 0.$$

Linearize f^2 by bounded divided differences in y and a vector divided difference in z :

$$d\Delta Y_t = -(\delta f_t + \alpha_t \Delta Y_t + \theta_t \cdot \Delta Z_t) dt + \Delta Z_t dB_t.$$

The process α is bounded, and (Q2) together with the boundedness of Y^1, Y^2 gives

$$|\theta_t| \leq C(1 + |Z_t^1| + |Z_t^2|).$$

The a priori estimate implies that $\int \theta dB$ is BMO. Define a probability measure $\mathbb{Q}$ by

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = \mathcal{E}_T(\theta).$$

Then $B_t^Q := B_t - \int_0^t \theta_s ds$ is a Brownian motion under Q . With $\Gamma_t := \exp(\int_0^t \alpha_s ds)$, localization followed by the reverse Hölder estimate gives

$$\Gamma_t \Delta Y_t = \mathbb{E}_t^Q \left[\Gamma_T (\xi_1 - \xi_2) + \int_t^T \Gamma_s \delta f_s ds \right] \leq 0.$$

Thus $\Delta Y_t \leq 0$. □

Lemma 5.2.2 (Monotone stability). Let (ξ_n, f^n) satisfy (Q1)–(Q3) uniformly, and let (Y^n, Z^n) be the corresponding solutions. Assume

$$\xi_n \rightarrow \xi \quad \text{in probability}, \quad f^n(\cdot, y, z) \rightarrow f(\cdot, y, z) \quad \text{in } dt \otimes d\mathbb{P} \text{ measure}$$

for each fixed (y, z) , and assume either $Y^n \uparrow Y$ or $Y^n \downarrow Y$. Then there is $Z \in \mathbb{H}^2(0, T; \mathbb{R}^d)$ such that (Y, Z) solves the limiting BSDE, and

$$\mathbb{E} \left[\int_0^T |Z_t^n - Z_t|^2 dt \right] \rightarrow 0.$$

Proof. We treat the increasing case. The a priori estimates give a uniform bound for Y^n in L^∞ and a uniform BMO bound for $\int Z^n dB$. In particular, (Z^n) is bounded in $\mathbb{H}^2$, so reflexivity gives a weakly convergent subsequence with limit Z . The BMO energy inequality also gives uniform integrability of

$$\left\{ \int_0^T |Z_t^n|^2 dt : n \geq 1 \right\}.$$

For $m > n$, apply Itô's formula to $\varphi(Y^m - Y^n)$, where

$$\varphi(y) := \frac{e^{4C_0 y} - 4C_0 y - 1}{8C_0^2}, \quad y \geq 0.$$

Then $\varphi(0) = \varphi'(0) = 0$ and $\varphi'' - 4C_0 \varphi' = 2$. The uniform quadratic-growth bound and Young's inequality give

$$\begin{aligned} \mathbb{E} \left[\int_0^T |Z_t^m - Z_t^n|^2 dt \right] &\leq C \mathbb{E} \left[\varphi(Y_T^m - Y_T^n) \right. \\ &\quad \left. + \int_0^T \varphi'(Y_t^m - Y_t^n) (1 + |Z_t^m|^2 + |Z_t^n|^2) dt \right]. \end{aligned}$$

Because the monotone sequence Y^n is uniformly bounded, $Y^m - Y^n \rightarrow 0$ pointwise as $m, n \rightarrow \infty$. Hence the first term and the factor $\varphi'(Y^m - Y^n)$ converge to zero. Uniform integrability of the energy variables makes the entire right-hand side tend to zero. Thus (Z^n) is Cauchy in $\mathbb{H}^2$, and its strong limit is the weak limit Z .

To pass to the generator, split

$$\begin{aligned} f_t^n(Y_t^n, Z_t^n) - f_t(Y_t, Z_t) &= f_t^n(Y_t^n, Z_t^n) - f_t^n(Y_t, Z_t) \\ &\quad + f_t^n(Y_t, Z_t) - f_t(Y_t, Z_t). \end{aligned}$$

The first term tends to zero in L^1 by the local Lipschitz estimate, bounded convergence of Y^n , strong H^2 convergence of Z^n , and the uniform BMO bounds. The second tends to zero in measure by approximation of (Y, Z) with simple processes and is uniformly integrable by the common quadratic-growth bound. It therefore converges in $L^1(dt \otimes d\mathbb{P})$. Passing to the limit in the BSDE proves the result. The decreasing case is analogous. $\square$

5.3 Well-posedness

Theorem 5.3.1 (Existence and uniqueness). Under (Q1)–(Q3), the quadratic BSDE admits a unique solution

$$(Y, Z) \in \mathbb{L}^\infty(\mathbb{F}; \mathbb{R}) \times H^2(0, T; \mathbb{R}^d).$$

The martingale $\int_0^\cdot Z_s dB_s$ is BMO.

Proof. Uniqueness follows by applying the comparison theorem in both directions. Once Y is fixed, uniqueness of Z follows from the BSDE and Itô isometry.

For existence, truncate the generator above and below:

$$f^n := f \wedge n, \quad f^{n,m} := f^n \vee (-m),$$

and regularize its z dependence by infimal convolution,

$$f_t^{n,m,k}(y, z) := \inf_{z' \in \mathbb{R}^d} (f_t^{n,m}(y, z') + k|z - z'|).$$

For fixed (n, m, k) , the generator is globally Lipschitz in (y, z) , so the Lipschitz BSDE has a unique solution $(Y^{n,m,k}, Z^{n,m,k})$. Comparison gives monotonicity in k , m , and n , while the quadratic a priori estimate gives the uniform bounds needed at each limiting step. First let $k \rightarrow \infty$, then $m \rightarrow \infty$, and finally $n \rightarrow \infty$. The monotone-stability lemma identifies the limits and produces a solution of the original equation. The BMO conclusion is the conditional Z estimate from the first theorem. $\square$

Theorem 5.3.2 (General stability). Let (ξ_n, f^n) , $n \geq 0$, satisfy (Q1)–(Q3) uniformly, and let (Y^n, Z^n) be the corresponding solutions. Suppose

$$\xi_n \rightarrow \xi_0 \quad \text{in probability}$$

and, for every fixed (y, z) ,

$$f^n(\cdot, y, z) \rightarrow f^0(\cdot, y, z) \quad \text{in } dt \otimes d\mathbb{P} \text{ measure.}$$

Then, for every $p \geq 1$,

$$\mathbb{E} \left[|(Y^n - Y^0)_T|^p + \left(\int_0^T |Z_t^n - Z_t^0|^2 dt \right)^{p/2} \right] \rightarrow 0.$$

Proof. Set $\Delta Y^n = Y^n - Y^0$ and $\Delta Z^n = Z^n - Z^0$. Linearization gives

$$d\Delta Y_t^n = -(\Delta f_t^n(Y_t^0, Z_t^0) + \alpha_t^n \Delta Y_t^n + \theta_t^n \cdot \Delta Z_t^n) dt + \Delta Z_t^n dB_t,$$

where α^n is uniformly bounded and

$$|\theta_t^n| \leq C(1 + |Z_t^n| + |Z_t^0|).$$

The BMO norms of $\int \theta^n dB$ are uniformly bounded. Girsanov's theorem and the uniform reverse Hölder inequality therefore give, for some $q > 1$ independent of n ,

$$|\Delta Y_t^n| \leq C \left(\mathbb{E}_t \left[\left(|\xi_n - \xi_0| + \int_t^T |\Delta f_s^n(Y_s^0, Z_s^0)| ds \right)^q \right] \right)^{1/q}.$$

The quantity on the right converges to zero in every finite L^r : uniform boundedness of the terminal conditions and the BMO exponential estimate provide uniform integrability of all required moments. This proves convergence of Y^n at fixed times.

Under the Girsanov measure, BDG applied to the linearized equation gives convergence of $\int |\Delta Z^n|^2$ in every finite moment. A small negative-moment estimate for the stochastic densities, again supplied by BMO analysis, transfers the estimate back to $\mathbb{P}$. Finally, the BSDE itself and BDG give convergence of the running supremum of ΔY^n . $\square$

Chapter 6 Forward–Backward Stochastic Differential Equations

We study fully coupled FBSDEs of the form

$$\begin{cases} X_t = x + \int_0^t b_s(X_s, Y_s, Z_s) \, ds + \int_0^t \sigma_s(X_s, Y_s, Z_s) \, dB_s, \\ Y_t = g(X_T) + \int_t^T f_s(X_s, Y_s, Z_s) \, ds - \int_t^T Z_s \, dB_s. \end{cases}$$

The processes X, Y, Z take values in $\mathbb{R}^{d_1}$, $\mathbb{R}^{d_2}$, and $\mathbb{R}^{d_2 \times d}$, respectively. We write $\Theta = (X, Y, Z)$ and

$$\|\Theta\|^2 := \mathbb{E} \left[\sup_{0 \leq t \leq T} |X_t|^2 + \sup_{0 \leq t \leq T} |Y_t|^2 + \int_0^T |Z_t|^2 \, dt \right].$$

Unless stated otherwise, assume:

(F1) $\mathbb{F} = \mathbb{F}^B$ is the usual augmented Brownian filtration;

(F2) with $\varphi_t^0 := \varphi_t(0, 0, 0)$ for $\varphi = b, \sigma, f$,

$$\begin{aligned} g(0) &\in L^2(\mathcal{F}_T; \mathbb{R}^{d_2}), \\ \mathbb{E} \left[\left(\int_0^T (|b_t^0| + |f_t^0|) \, dt \right)^2 + \int_0^T |\sigma_t^0|^2 \, dt \right] &< \infty; \end{aligned}$$

(F3) b, σ, f , and g are uniformly Lipschitz in their spatial variables.

6.1 Local well-posedness by a contraction argument

Let $L_{\sigma, z}$ denote the Lipschitz constant of σ in z and let L_g be the Lipschitz constant of g .

Theorem 6.1.1 (Small-time well-posedness). Assume (F1)–(F3) and

$$L_{\sigma, z} L_g < 1.$$

There are constants $\delta_0 > 0$ and $C < \infty$, depending only on the Lipschitz constants, the dimensions, and the product $L_{\sigma, z} L_g$, such that, whenever $T \leq \delta_0$, the FBSDE has a unique solution and

$$\|\Theta\|^2 \leq C(|x|^2 + I_0^2),$$

where

$$I_0^2 := \mathbb{E} \left[\left(\int_0^T (|b_t^0| + |f_t^0|) dt \right)^2 + \int_0^T |\sigma_t^0|^2 dt + |g(0)|^2 \right].$$

Proof. We present the scalar proof; the multidimensional proof differs only in notation. For progressively measurable input processes $(\mathbf{y}, \mathbf{z})$, first solve

$$\begin{aligned} X_t^{\mathbf{y}, \mathbf{z}} &= x + \int_0^t b_s(X_s^{\mathbf{y}, \mathbf{z}}, \mathbf{y}_s, \mathbf{z}_s) ds \\ &\quad + \int_0^t \sigma_s(X_s^{\mathbf{y}, \mathbf{z}}, \mathbf{y}_s, \mathbf{z}_s) dB_s, \end{aligned}$$

and then solve the Lipschitz BSDE

$$\begin{aligned} Y_t^{\mathbf{y}, \mathbf{z}} &= g(X_T^{\mathbf{y}, \mathbf{z}}) + \int_t^T f_s(X_s^{\mathbf{y}, \mathbf{z}}, Y_s^{\mathbf{y}, \mathbf{z}}, Z_s^{\mathbf{y}, \mathbf{z}}) ds \\ &\quad - \int_t^T Z_s^{\mathbf{y}, \mathbf{z}} dB_s. \end{aligned}$$

Define $F(\mathbf{y}, \mathbf{z}) := (Y^{\mathbf{y}, \mathbf{z}}, Z^{\mathbf{y}, \mathbf{z}})$ on the Banach space with norm

$$\|(\mathbf{y}, \mathbf{z})\|_w^2 := \sup_{0 \leq t \leq T} \mathbb{E} |\mathbf{y}_t|^2 + \mathbb{E} \left[\int_0^T |\mathbf{z}_t|^2 dt \right].$$

For two inputs, use Δ for differences. Linearization gives

$$\begin{aligned} \Delta X_t &= \int_0^t (\alpha_s^1 \Delta X_s + \beta_s^1 \Delta \mathbf{y}_s + \gamma_s^1 \Delta \mathbf{z}_s) ds \\ &\quad + \int_0^t (\alpha_s^2 \Delta X_s + \beta_s^2 \Delta \mathbf{y}_s + \gamma_s^2 \Delta \mathbf{z}_s) dB_s, \\ \Delta Y_t &= \lambda \Delta X_T + \int_t^T (\alpha_s^3 \Delta X_s + \beta_s^3 \Delta Y_s + \gamma_s^3 \Delta Z_s) ds - \int_t^T \Delta Z_s dB_s, \end{aligned}$$

where the coefficients are bounded, $|\gamma^2| \leq L_{\sigma, \mathbf{z}}$, and $|\lambda| \leq L_g$. For $0 < \varepsilon < 1/2$, Itô's formula and Young's inequality yield, provided $CT/\varepsilon \leq \varepsilon$,

$$\sup_{t \leq T} \mathbb{E} |\Delta X_t|^2 \leq \frac{L_{\sigma, \mathbf{z}}^2 + 2\varepsilon}{1 - \varepsilon} \|(\Delta \mathbf{y}, \Delta \mathbf{z})\|_w^2$$

and

$$\|(\Delta Y, \Delta Z)\|_w^2 \leq \frac{L_g^2 + \varepsilon^2}{1 - 2\varepsilon} \sup_{t \leq T} \mathbb{E} |\Delta X_t|^2.$$

Consequently,

$$\|(\Delta Y, \Delta Z)\|_w^2 \leq c_\varepsilon \|(\Delta \mathbf{y}, \Delta \mathbf{z})\|_w^2,$$

where

$$c_\varepsilon := \frac{(L_g^2 + \varepsilon^2)(L_{\sigma, \mathbf{z}}^2 + 2\varepsilon)}{(1 - \varepsilon)(1 - 2\varepsilon)}.$$

As $\varepsilon \downarrow 0$, $c_\varepsilon \rightarrow (L_g L_{\sigma, \mathbf{z}})^2 < 1$. Choose ε so that $c_\varepsilon < 1$, and then choose δ_0 so that $CT/\varepsilon \leq \varepsilon$ whenever $T \leq \delta_0$. Banach's fixed point theorem gives a unique fixed point (Y, Z) ; with $X = X^{Y, Z}$, the triple (X, Y, Z) is the unique solution of the FBSDE.

For the estimate, let $(Y^0, Z^0) = F(0, 0)$. The contraction property gives

$$\|(Y, Z)\|_w \leq \frac{1}{1 - \sqrt{c_\varepsilon}} \|(Y^0, Z^0)\|_w.$$

Standard SDE and BSDE estimates give $\|(Y^0, Z^0)\|_w \leq C(|x| + I_0)$, and a final SDE estimate for X proves the asserted bound for Θ . $\square$

Corollary 6.1.2 (Local stability). Let two data sets satisfy the hypotheses of the theorem with the same local horizon. Denote their solutions by Θ and $\tilde{\Theta}$, and write $\Delta\varphi = \tilde{\varphi} - \varphi$. Then

$$\begin{aligned} \|\tilde{\Theta} - \Theta\|^2 \leq \mathbb{C}\mathbb{E} & \left[|\tilde{x} - x|^2 + \left(\int_0^T (|\Delta b_t(\tilde{\Theta}_t)| + |\Delta f_t(\tilde{\Theta}_t)|) dt \right)^2 \right. \\ & \left. + \int_0^T |\Delta \sigma_t(\tilde{\Theta}_t)|^2 dt + |\Delta g(\tilde{X}_T)|^2 \right]. \end{aligned}$$

Remark. The condition $L_{\sigma, z} L_g < 1$ is sufficient, not necessary. For example,

$$X_t = x + \gamma \int_0^t Z_s dB_s, \quad Y_t = \lambda X_T - \int_t^T Z_s dB_s$$

is well posed if and only if $\gamma\lambda \neq 1$.

6.2 The decoupling-field method

On a local interval $[t, T]$, let $\Theta^{t,x}$ be the solution starting from $X_t^{t,x} = x$, and define

$$u_t(x) := Y_t^{t,x}.$$

For random coefficients, u is generally a random field.

Theorem 6.2.1 (Local decoupling field). Under the hypotheses of the small-time theorem:

- (i) $u_t(x) \in L^2(\mathcal{F}_t; \mathbb{R}^{d_2})$ for every (t, x) ;
- (ii) there is a deterministic constant C such that, outside a single null set,

$$|u_t(x_1) - u_t(x_2)| \leq C|x_1 - x_2|$$

for all t, x_1, x_2 ;

- (iii) every local solution satisfies

$$Y_s^{t,x} = u_s(X_s^{t,x}), \quad t \leq s \leq T.$$

Proof. Measurability follows from the construction of the local solution. The conditional version of the local stability estimate gives the pathwise Lipschitz inequality. Finally, uniqueness on the interval $[s, T]$ identifies the continuation of the solution from time s with the solution starting at $(s, X_s^{t,x})$, proving the flow identity. $\square$

Definition 6.2.2. An $\mathbb{F}$ -measurable random field

$$u : [0, T] \times \mathbb{R}^{d_1} \times \Omega \longrightarrow \mathbb{R}^{d_2}, \quad u_T(x) = g(x),$$

is a decoupling field if there exists $\delta > 0$ such that, for every $t_1 < t_2$ with $t_2 - t_1 \leq \delta$ and every $\eta \in L^2(\mathcal{F}_{t_1}; \mathbb{R}^{d_1})$, the FBSDE on $[t_1, t_2]$ with $X_{t_1} = \eta$ and terminal condition $Y_{t_2} = u_{t_2}(X_{t_2})$ has a unique solution satisfying

$$Y_t = u_t(X_t), \quad t_1 \leq t \leq t_2.$$

The decoupling field is regular if it is uniformly Lipschitz continuous in x .

Remark. A decoupling field is unique. Indeed, on a sufficiently short interval, start at an arbitrary deterministic state x at its left endpoint. Local uniqueness identifies the value of $u_t(x)$. Repeating this argument backward over a time partition identifies the field on the whole interval.

Theorem 6.2.3 (Global solution from a decoupling field). If the FBSDE has a decoupling field u , then it has a unique global solution, and

$$Y_t = u_t(X_t), \quad 0 \leq t \leq T.$$

Proof. Choose a partition $0 = t_0 < \dots < t_N = T$ with mesh at most the local length in the definition. Starting with $X_0 = x$, solve successively on $[t_{i-1}, t_i]$ the FBSDE with terminal condition $u_{t_i}(X_{t_i})$. The identities $Y_{t_i} = u_{t_i}(X_{t_i})$ make the pieces agree at the grid points, so they paste into a global solution. For uniqueness, restrict any global solution to the final interval, use local uniqueness there, and proceed backward interval by interval. $\square$

6.2.1 A structural condition for global regularity

Theorem 6.2.4. Assume $d = d_1 = d_2 = 1$, σ is independent of z , the coefficients are continuously differentiable in (x, y, z) with bounded derivatives, and there is a constant $c > 0$ such that

$$\partial_y \sigma \partial_z b \leq -c |\partial_y b + \partial_x \sigma \partial_z b + \partial_y \sigma \partial_z f|.$$

Then the FBSDE has a regular decoupling field and hence a unique global solution.

Proof. Local well-posedness holds because $\partial_z \sigma = 0$. Consider a local solution on $[t_0, t_1]$ with terminal condition $Y_{t_1} = h(X_{t_1})$, where h is Lipschitz. Differentiate the FBSDE with respect to the initial state and write $\nabla \Theta = (\nabla X, \nabla Y, \nabla Z)$. Before a possible zero of ∇X , define

$$\hat{Y} := \frac{\nabla Y}{\nabla X}, \quad \hat{Z} := \frac{\nabla Z}{\nabla X} - \hat{Y}(\partial_x \sigma + \partial_y \sigma \hat{Y}).$$

Itô's formula gives the scalar BSDE

$$d\hat{Y}_s = -(\partial_x f_s + a_s(\hat{Y}_s)\hat{Y}_s + \theta_s(\hat{Y}_s)\hat{Z}_s) ds + \hat{Z}_s dB_s,$$

with terminal value $h'(X_{t_1})$, where

$$\begin{aligned} a(y) &:= \partial_y f + \partial_x b + \partial_z f \partial_x \sigma \\ &\quad + (\partial_y b + \partial_z f \partial_y \sigma + \partial_z b \partial_x \sigma) y + \partial_z b \partial_y \sigma y^2, \\ \theta(y) &:= \partial_z f + \partial_x \sigma + (\partial_z b + \partial_y \sigma) y. \end{aligned}$$

The forward variational equation can be written as a stochastic exponential, which proves that ∇X never reaches zero. The structural assumption and completion of the square give a deterministic upper bound

$$a(y) \leq \bar{L} \quad \text{for all } y,$$

where $\bar{L}$ depends only on the derivative bounds and c . First take a smooth terminal map h and stop when $|\hat{Y}| + |\nabla X| + |\nabla X|^{-1}$ reaches a fixed level. On the stopped interval, $\theta(\hat{Y})$ is bounded, so Girsanov's theorem and the linear representation of the characteristic BSDE give

$$|\hat{Y}_t| \leq \|h'\|_\infty e^{\bar{L}(t_1-t)} + L \int_t^{t_1} e^{\bar{L}(s-t)} ds.$$

The bound is independent of the stopping level; hence the stopping times increase to t_1 . A smooth approximation of a merely Lipschitz terminal map gives the same estimate with $\|h'\|_\infty$ replaced by the Lipschitz constant of h . Thus the Lipschitz constant of the decoupling field on a local interval is controlled solely by the Lipschitz constant at the right endpoint and fixed model constants. Iterating this estimate backward from $h = g$ gives a finite bound on the whole horizon. The local existence length can therefore be chosen uniformly, so the decoupling field extends to time zero. The preceding theorem then gives the global solution. $\square$

6.3 Properties derived from the decoupling field

Theorem 6.3.1 (Comparison of decoupling fields). Let (b, σ, f^i, g^i) , $i = 1, 2$, satisfy the hypotheses of Theorem 6.2.4, and let u^i be their regular decoupling fields. If

$$f^1 \leq f^2 \quad \text{and} \quad g^1 \leq g^2,$$

then $u^1 \leq u^2$.

Proof. It is enough to work on one common local interval and then iterate backward. Fix the same initial state for the two FBSDEs and linearize the difference. The terminal difference has the form

$$\Delta Y_T = \lambda \Delta X_T + \Delta g(X_T^1), \quad \Delta g := g^1 - g^2 \leq 0,$$

and the generator difference is the sum of a linear term in $(\Delta X, \Delta Y, \Delta Z)$ and $\Delta f(\Theta^1)$, where $\Delta f = f^1 - f^2 \leq 0$.

Solve the associated homogeneous linear FBSDE with unit initial forward state and terminal coefficient λ . Its forward component is strictly positive. Dividing its backward component by its forward component produces bounded processes $(\hat{Y}, \hat{Z})$. Set

$$\delta Y := \Delta Y - \hat{Y} \Delta X$$

and choose the corresponding δZ so that the coefficient of ΔX cancels. Then $(\delta Y, \delta Z)$ solves a scalar linear BSDE with terminal value $\Delta g(X_T^1) \leq 0$ and inhomogeneous term $\Delta f(\Theta^1) \leq 0$. The linear representation formula, after the standard BMO localization for its stochastic exponential, gives $\delta Y_0 \leq 0$. Since $\Delta X_0 = 0$, this is $u_0^1(x) - u_0^2(x) \leq 0$. Applying the same argument from each starting time and then iterating over the partition proves the result. $\square$

Remark. This theorem compares the decoupling fields, not the two backward processes along their different forward states. In general, $X_t^1 \neq X_t^2$, so no pointwise comparison between $Y_t^1 = u_t^1(X_t^1)$ and $Y_t^2 = u_t^2(X_t^2)$ follows for $t > 0$.

Theorem 6.3.2 (A variational estimate). Consider the scalar linear FBSDE

$$\begin{cases} X_t = 1 + \int_0^t (\alpha_s^1 X_s + \beta_s^1 Y_s + \gamma_s^1 Z_s) ds + \int_0^t (\alpha_s^2 X_s + \beta_s^2 Y_s) dB_s, \\ Y_t = GX_T + \int_t^T (\alpha_s^3 X_s + \beta_s^3 Y_s + \gamma_s^3 Z_s) ds - \int_t^T Z_s dB_s. \end{cases}$$

Assume $|\alpha^i|, |\beta^i|, |\gamma^i| \leq K$, $|G| \leq K_0$, and

$$\beta^2 \gamma^1 = 0, \quad \beta^1 + \alpha^2 \gamma^1 + \beta^2 \gamma^3 = 0.$$

For a sufficiently small horizon, the FBSDE has a unique solution and

$$|Y_t| \leq \bar{K}_0 |X_t|, \quad \bar{K}_0 := (K_0 + 1)e^{(2K+K^2)T} - 1.$$

In particular, $|Y_0| \leq \bar{K}_0$.

Proof. Local well-posedness follows from the small-time theorem. Conditional stability, tested against arbitrary bounded $\mathcal{F}_t$ -measurable multipliers, first gives $|Y_t| \leq C|X_t|$. Before the first zero of X , set

$$\tilde{Y} := Y/X, \quad \tilde{Z} := Z/X - \tilde{Y}(\alpha^2 + \beta^2 \tilde{Y}).$$

The two structural equalities eliminate the quadratic and linear powers of $\tilde{Y}$ from the drift coefficient generated by Itô's formula. After localization at the levels $X = 1/n$ and a Girsanov change of measure, the linear representation gives

$$|\tilde{Y}_t| \leq K_0 e^{(2K+K^2)(T-t)} + K \int_t^T e^{(2K+K^2)(s-t)} ds \leq \bar{K}_0.$$

The stochastic-exponential representation of X shows that it cannot hit zero before the terminal

time; the localization can therefore be removed. Multiplying by $|X_t|$ proves the estimate. $\square$

6.4 The method of continuation

We now return to scalar FBSDEs and impose the monotonicity conditions: there is $c > 0$ such that, for every $\theta_i = (x_i, y_i, z_i)$ and $\Delta\theta = \theta_1 - \theta_2$,

$$\begin{aligned} & [b_t(\theta_1) - b_t(\theta_2)]\Delta y + [\sigma_t(\theta_1) - \sigma_t(\theta_2)]\Delta z \\ & - [f_t(\theta_1) - f_t(\theta_2)]\Delta x \leq -c(|\Delta x|^2 + |\Delta y|^2 + |\Delta z|^2), \\ & [g(x_1) - g(x_2)]\Delta x \geq 0. \end{aligned}$$

Theorem 6.4.1 (Uniqueness under monotonicity). Under (F1)–(F3) and the monotonicity conditions, the FBSDE has at most one solution.

Proof. Let Θ^1 and Θ^2 be two solutions and use Δ for differences. Itô's formula gives

$$\begin{aligned} d(\Delta X_t \Delta Y_t) &= (-\Delta f_t \Delta X_t + \Delta b_t \Delta Y_t + \Delta \sigma_t \Delta Z_t) dt \\ &\quad + (\Delta X_t \Delta Z_t + \Delta \sigma_t \Delta Y_t) dB_t. \end{aligned}$$

Since $\Delta X_0 = 0$ and $\Delta Y_T = g(X_T^1) - g(X_T^2)$, expectation and monotonicity yield

$$0 \leq -c\mathbb{E} \left[\int_0^T (|\Delta X_t|^2 + |\Delta Y_t|^2 + |\Delta Z_t|^2) dt \right].$$

Thus $\Delta Z = 0$ almost everywhere and, by continuity of X and Y , $\Delta X = \Delta Y = 0$ for every time. $\square$

Lemma 6.4.2 (Solvable reference equation). Let b^0, f^0 be progressively measurable with

$$\mathbb{E} \left[\left(\int_0^T (|b_t^0| + |f_t^0|) dt \right)^2 \right] < \infty,$$

let $\sigma^0 \in \mathbb{H}^2(0, T)$, and let $g^0 \in L^2(\mathcal{F}_T)$. Then

$$\begin{cases} X_t = x + \int_0^t (-Y_s + b_s^0) ds + \int_0^t (-Z_s + \sigma_s^0) dB_s, \\ Y_t = X_T + g^0 + \int_t^T (X_s + f_s^0) ds - \int_t^T Z_s dB_s \end{cases}$$

has a unique solution.

Proof. Uniqueness follows from the monotonicity estimate with $c = 1$. For existence, set $\bar{Y} := Y - X$. Direct subtraction of the forward and backward differentials shows that

$$\bar{Y}_t = g^0 + \int_t^T (-\bar{Y}_s + f_s^0 + b_s^0) ds - \int_t^T \bar{Z}_s dB_s,$$

where

$$\bar{Z} = 2Z - \sigma^0.$$

This linear BSDE has a unique solution. Set

$$Z := \frac{1}{2}(\bar{Z} + \sigma^0)$$

and solve

$$X_t = x + \int_0^t (-X_s - \bar{Y}_s + b_s^0) ds + \frac{1}{2} \int_0^t (\sigma_s^0 - \bar{Z}_s) dB_s.$$

Finally, define $Y = X + \bar{Y}$. Substitution verifies the original FBSDE. $\square$

For $\alpha \in [0, 1]$, define

$$b_t^\alpha(x, y, z) := \alpha b_t(x, y, z) - (1 - \alpha)y,$$

$$\sigma_t^\alpha(x, y, z) := \alpha \sigma_t(x, y, z) - (1 - \alpha)z,$$

$$f_t^\alpha(x, y, z) := \alpha f_t(x, y, z) + (1 - \alpha)x,$$

$$g^\alpha(x) := \alpha g(x) + (1 - \alpha)x.$$

These coefficients satisfy the monotonicity condition with constant

$$c_\alpha := \alpha c + 1 - \alpha \geq \min(c, 1).$$

Let $\text{FBSDE}(\alpha)$ denote the equation with coefficients $(b^\alpha, \sigma^\alpha, f^\alpha, g^\alpha)$ and arbitrary additive data $(b^0, \sigma^0, f^0, g^0)$ in the spaces used in the preceding lemma. Call $\text{FBSDE}(\alpha)$ solvable if it has a solution for every such additive data set.

Lemma 6.4.3 (Continuation step). If $\text{FBSDE}(\alpha_0)$ is solvable, then there is a number $\delta_0 > 0$, depending only on the common Lipschitz and monotonicity constants, such that $\text{FBSDE}(\alpha)$ is solvable for every

$$\alpha \in [\alpha_0, \min(1, \alpha_0 + \delta_0)].$$

Proof. Write $\delta = \alpha - \alpha_0$. Starting from $\Theta^0 = 0$, define Θ^{n+1} as the solution of $\text{FBSDE}(\alpha_0)$ with additive data

$$b_t^{n,0} := b_t^0 + \delta(Y_t^n + b_t(\Theta_t^n)),$$

$$\sigma_t^{n,0} := \sigma_t^0 + \delta(Z_t^n + \sigma_t(\Theta_t^n)),$$

$$f_t^{n,0} := f_t^0 + \delta(f_t(\Theta_t^n) - X_t^n),$$

$$g^{n,0} := g^0 + \delta(g(X_T^n) - X_T^n).$$

Solvability at α_0 makes the iteration well defined. Let $\Delta\Theta^n = \Theta^{n+1} - \Theta^n$. Itô's formula for $\Delta X^n \Delta Y^n$, the monotonicity of the α_0 coefficients, and Young's inequality control the natural L^2 norm of $\Delta\Theta^n$ by the preceding increment. The terminal increment

$$\delta(g(X_T^n) - g(X_T^{n-1}) - \Delta X_T^{n-1})$$

must be retained in this estimate; it is absorbed using the Lipschitz bound for g and the standard SDE estimate. Altogether,

$$\|\Delta\Theta^n\|_{L^2(dt \otimes d\mathbb{P})}^2 \leq C\delta^2 \|\Delta\Theta^{n-1}\|_{L^2(dt \otimes d\mathbb{P})}^2.$$

The terms of order δ multiplying the current increment can be absorbed into the monotonicity constant. Choose δ_0 so that $C\delta_0^2 < 1/4$. Then the iteration is Cauchy. Passing to the limit in the Lipschitz coefficients shows that its limit solves $\text{FBSDE}(\alpha_0 + \delta)$. $\square$

Theorem 6.4.4 (Global well-posedness by continuation). Under (F1)–(F3) and the monotonicity conditions, the FBSDE admits a unique solution on every finite horizon.

Proof. The reference lemma proves solvability at $\alpha = 0$. Choose an integer N such that

$$(N - 1)\delta_0 < 1 \leq N\delta_0.$$

Applying the continuation-step lemma at most N times reaches $\alpha = 1$, which is the original FBSDE. Uniqueness was proved above. $\square$

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